Rights, Genomes, and Algorithms

An Integrated Portfolio of Constitutional Jurisprudence, Genomic Data Sovereignty, Semantic Computing, and Post-Quantum Public Infrastructure

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This volume consolidates and supersedes the earlier Rights, Genomes, and Algorithms *monograph and incorporates the HSPA v3.0 / SSE-9 / RIGOR-PRS Master Technical Specification.*

Abstract

This book consolidates a decade of work by a single author into one continuous argument. It moves from constitutional litigation, through the governance of genomic and neurophysiological data, into the design of a semantic-computing architecture and a post-quantum trust chain, and finally into a proposed public financing and treaty framework. The unifying claim is that the same structural weakness — the absence of enforceable constraint on institutions that process personal data — appears simultaneously as a civil-rights injury, as a genomic-privacy defect, as an algorithmic-governance problem, and as a cryptographic and financial one.

The volume has three audiences. For the legal reader it supplies a doctrine-by- doctrine map of actionable claims under 42 U.S.C. § 1983, the statutory lattice of GINA, the ADA, and HIPAA, and the evidentiary standards that govern neurogenetic proof. For the technical reader it reconstructs the Hybrid Semantic Processing Architecture (HSPA v3.0) — its Semantic Execution Graph, its probabilistic control layer, its reconfigurable compute fabric, its nitrogen-vacancy quantum substrate, and its ML-KEM/ML-DSA/SLH-DSA trust chain — and separates what is standardised, what is simulated, and what remains a design target. For the policy reader it sets out an international framework that joins cognitive accessibility, genomic justice, and quantum-secure infrastructure into a single directive.

Throughout, the book maintains a deliberate distinction between four registers of statement: adopted law, legislative or architectural proposal, model and simulation, and design target. That discipline is not a stylistic ornament; it is the central methodological contribution of this edition.

Status and Provenance Notice

A reader who encounters litigation theory, a quantum substrate, a trust indenture, and a United Nations petition inside one cover is entitled to know which claims rest on enactment, which on doctrine, which on computation, and which on aspiration. This edition therefore labels its material.

[ADOPTED LAW]
Binding law or a published standard. Examples: the Administrative Procedure Act; 42 U.S.C. § 1983; the Genetic Information Nondiscrimination Act of 2008; NIST FIPS 203/204/205.

[PROPOSAL]
A legislative, architectural, or organisational design that the author advocates but that no legislature or standards body has adopted. Examples: the Gov-LLM concept; the PSDLEF Sovereign Exchange Trust; the proposed UN commission of inquiry.

[MODEL / SIMULATION]
A computational artefact — a simulation, a reference implementation, or a numerical result produced by the author's own code. Examples: the MORTAR triage engine; the VQE validation of the quantum substrate; the DYREO allocation optimiser.

[DESIGN TARGET]
An engineering or operational goal not yet demonstrated at the scale claimed. Examples: the 600 MW/1 GWh solar-and-battery microgrid; the HARMONI-X cognitive prosthetic; the quantum gravitational reactor.

Where a factual assertion depends on a court docket, an administrative record, or an experimental measurement that the author has not yet placed in the reviewing record, the text carries the marker **[CITE REQUIRED]** and identifies the document that would satisfy it. This convention is inherited from the source portfolio's own working drafts, in which procedural assertions are systematically conditioned on file-stamped proof. It is preserved here because a manuscript that mixes advocacy with verifiable record must be readable as a record.

Two claims that recur and must not be conflated

The normative claim. Existing doctrine under-protects genomic, neurophysiological, and cognitive data, and that under-protection is remediable by identifiable legal and technical means. This claim is argued from primary authority and is open to ordinary legal dispute.

The engineering claim. A specific architecture — HSPA v3.0 — can deliver the privacy, accessibility, and auditability properties the normative claim requires. This claim is supported by simulation and reference implementation, not by fabricated silicon; the reader should treat performance figures as modelled unless a measurement is expressly identified.

A Note on Sources and Method

The material in this book derives from two bodies of primary text. The first is the author's project archive: several hundred files containing pleadings, portfolio volumes, patent disclosures, administrative correspondence, and technical notes. The second is the *HSPA v3.0 / SSE-9 / RIGOR-PRS Master Technical Specification*, a consolidated engineering document that supplies the architecture described in Part IV and the trust model described in Parts VI and VIII.

Three editorial operations were performed on that material.

1. **Consolidation.** Repetitions across portfolio volumes were eliminated; the several drafts of the same petition, report, or claim chart were reconciled into single authoritative texts. Where two drafts disagreed, the more complete and the more recent was preferred, and the discrepancy is noted.
2. **Sourcing.** Every quantitative claim, legal standard, and technical parameter was checked against an external authority. Where the source material asserted a figure without authority, a citation was supplied if one exists (for example, the Government Accountability Office's disability-appeal mortality accounting, the NIST post-quantum standards, or the locality-sensitive hashing literature) and a **[CITE REQUIRED]** marker if none does.
3. **Registering.** Each substantive proposition was assigned to one of the four registers described above, and the prose was rewritten so that the register is visible to a reader who does not read the labels.

The references are collected in the consolidated bibliography at the end of the volume. Primary authority — statutes, regulations, and reported decisions — is cited in the conventional form so that it can be verified independently. The author's own computational results are cited to the artefacts that produced them, and are labelled as **[MODEL / SIMULATION]** throughout.

Reader's Guide

The book is organised so that it can be read in three ways.

As a whole.
Parts I–II establish the legal injury and the constitutional argument; Parts III–VI supply the scientific and technical remedy; Parts VII–VIII situate the remedy in international law and in a financing structure; Part IX confronts the record and its gaps; Parts X–XII generalise the result into policy and close the argument.

As a technical reference.
Parts IV–VI are self-contained. A reader who wants only the architecture can begin at Chapter 11 and follow the six-layer stack to the verification chapter.

As a litigation toolkit.
Parts II and IX are self-contained. Part II states the claims and their doctrinal support; Part IX states what the record does and does not yet prove.

Chapter 2 is the hinge. It sets out the status discipline that governs every later chapter, and a reader who internalises it will be able to place any assertion in the book on one of the four registers without consulting the labels.

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Part I

Executive Synthesis: The Unified Systemic Framework

CHAPTER 1

The Integrated Portfolio: Scope, Structure, and Central Claim

The several parts of this work are not a miscellany. They are the same argument, made once in the language of doctrine, once in the language of engineering, and once in the language of finance.

1.1 What this book is

This is a single-author monograph that joins four normally separate literatures. The first is constitutional and civil-rights litigation. The second is the governance of genomic and neurophysiological data. The third is the design of secure, accessible computing infrastructure. The fourth is the financing of public-benefit digital infrastructure through trust and treaty instruments.

Its central claim is structural. Institutions that collect personal data — police departments, disability adjudicators, hospital systems, insurers, and agencies that deploy machine-learning models — are constrained today chiefly by statutes that were drafted before the data existed in its present form. The result is a persistent gap between the sensitivity of the data and the enforceability of the duty owed to the person the data describes. That gap appears as a civil-rights violation in the courthouse, as an evidentiary problem in the disability hearing, as an accuracy and bias problem in the algorithmic system, and as a cryptographic exposure in the data centre. The book treats these as four faces of one defect, and it proposes doctrine, evidence, and infrastructure to close it.

1.2 The five domains

The portfolio falls into five domains, each of which is developed in its own part and each of which is anchored by at least one primary artefact.

Constitutional litigation.
A five-count complaint under 42 U.S.C. § 1983, together with the appellate and extraordinary-writ practice needed to preserve it. The governing questions are the retaliation exception of *Nieves v. Bartlett*, municipal liability under *Monell*, and the abstention and immunity doctrines that shape the pleading.

Genomic and neurogenetic data.
A constitutional case for treating the genome and its neurophysiological correlates as information deserving heightened protection, the statutory lattice that partially supplies it, and the evidentiary

standard under which a polygenic or imaging biomarker may enter a courtroom or an administrative record.

Semantic computing.
The Hybrid Semantic Processing Architecture (HSPA v3.0), a six-layer system spanning a semantic execution graph, a probabilistic control layer, a reconfigurable compute fabric, a nitrogen-vacancy quantum substrate, a post-quantum trust chain, and a cognitive-accessibility interface.

Algorithmic governance.
The MORTAR triage engine and the OpenSSA integrity pipeline — instruments for making administrative prioritisation auditable and for keeping an evidentiary chain intact from capture to adjudication.

Public finance and treaty.
The PSDLEF Sovereign Exchange Trust, a proposed perpetual-benefit structure, together with the capital-allocation optimiser (DYREO), the algorithm execution suite (ADR-31), and the international policy framework that would give the whole structure a legal home.

1.3 How the parts fit together

Part II states the injury and the doctrine. Part III supplies the scientific evidence that the injury is real and measurable. Parts IV–VI supply the technical machinery that would let an institution process the same data without reproducing the injury. Part VII and Part VIII give the remedy a legal and a financial form. Part IX is the reckoning: the statement of facts and procedural history, written in the register of a court filing, in which every assertion is either supported by a record citation or plainly marked as requiring one. Parts X–XII generalise and close.

Existing law under-protects cognitive, genomic, and neurophysiological data; that under-protection is measurable as harm, remediable as doctrine, mitigable as architecture, and financeable as a perpetual-benefit trust.

1.4 A note on scale and honesty

The portfolio is large, and scale invites inflation. Several components of the engineering argument are simulations, not measurements; several components of the financial argument are proposals, not instruments; several components of the litigation argument are drafting plans, not filed pleadings. Chapter 2 states the discipline that the rest of the book follows in marking these distinctions, and Part IX applies it most severely to the author's own record.

CHAPTER 2

Status Discipline: Proposal, Prototype, Simulation, and Adopted Law

> The first duty of a document that mixes advocacy with proof is to say which is which.

2.1 Why a register taxonomy is needed

A reader coming to this material from any single discipline will find something in it that looks strange. A lawyer will find a quantum substrate. An engineer will find a complaint under § 1983. A policy analyst will find a trust indenture. The risk is not that any one part is wrong; it is that the reader will transfer the confidence appropriate to one register onto material that belongs to another. The remedy is an explicit taxonomy, applied without exception.

2.2 The four registers

Principle 2.1 (Register separation)**.** *Every substantive proposition in this book belongs to exactly one of four registers: adopted law, proposal, model, or target. A proposition that has not been assigned to a register is not yet ready to be relied upon.*

Register one: adopted law. A rule that binds, or a standard that has been published by a competent body. Examples used throughout the book include the Administrative Procedure Act; 42 U.S.C. § 1983; the Genetic Information Nondiscrimination Act of 2008; the Americans with Disabilities Act and its 2008 amendments; the Health Insurance Portability and Accountability Act and its Security Rule; and the post-quantum standards published by the National Institute of Standards and Technology in August 2024 as FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA). Adopted law is cited in conventional form and can be verified independently.

Register two: proposal. A design that the author advocates and that no authority has adopted. The Gov-LLM concept — a government large language model operated as a reasonable accommodation — is a proposal. The PSDLEF Sovereign Exchange Trust is a proposal. The request for a United Nations commission of inquiry is a proposal. Proposals are argued, not assumed; their legal feasibility is examined rather than asserted.

Register three: model and simulation. A computational result. When this book reports that a hashing scheme achieves a given recall at a given precision, or that an optimiser allocates capital in a given ratio, or

that a variational eigensolver converges to an energy within a stated tolerance, the figure comes from the author's own code. It is reproducible from the reference implementations, and it is a statement about the model rather than about a deployed system.

Register four: design target. An engineering goal not yet demonstrated at the scale claimed. The 600 MW/1 GWh solar-and-battery microgrid, the HARMONI-X cognitive prosthetic, and the quantum gravitational reactor are design targets. They are described in the terms a specification would use, because that is the useful way to describe them, but they are not reported as achievements.

2.3 Worked examples of the discipline

The taxonomy is easiest to see in borderline cases.

- *Post-quantum cryptography.* That ML-KEM-768 uses a 1,184-byte encapsulation key and a 1,088-byte ciphertext is **[ADOPTED LAW]** — it is fixed by FIPS 203 and by the IETF's ML-KEM profile. That the author's trust chain selects ML-KEM-1024 for a given data class is **[PROPOSAL]**. That the chain's end-to-end latency under load is some particular number is **[MODEL / SIMULATION]**.
- *Disability adjudication.* That the Social Security Administration processes appeals with substantial delay is **[ADOPTED LAW]** in the sense that it is documented by the Government Accountability Office. That a locality-sensitive-hashing triage engine would reduce that delay to a stated figure is **[MODEL / SIMULATION]**. That the agency should adopt such an engine is **[PROPOSAL]**.
- *Genomic evidence.* That a polygenic score is admissible only if it survives the framework of *Daubert* is **[ADOPTED LAW]**. That a particular score meets the threshold is a **[MODEL / SIMULATION]** claim about the author's data. That courts should adopt a specific disclosure protocol for scores is **[PROPOSAL]**.

The trigger for a citation marker

Where a proposition depends on a fact that lives in a court docket, an administrative record, or a laboratory measurement, and that fact has not been placed in the reviewing record, the text carries **[CITE REQUIRED]** and names the document that would close the gap. This convention runs through Part IX, where it does the most work.

2.4 The consequence for the reader

The discipline has a practical payoff. A reader who wants to know whether a sentence can be acted upon — cited in a brief, implemented in hardware, funded by an investor — can answer the question from the register alone. The rest of the book applies the taxonomy silently, in prose written so that the register is apparent even to a reader who ignores the labels.

Part II

Constitutional Jurisprudence and Biological Data Sovereignty

CHAPTER 3

The Five Counts of the § 1983 Complaint

Section 1983 does not create duties. It creates a remedy for duties created elsewhere — and the whole art of the pleading lies in naming the duty the defendant is said to have broken.

3.1 The statutory premise

Section 1983 of Title 42 imposes liability on any person who, under colour of state law, subjects another to the deprivation of a right secured by the Constitution or laws of the United States. The provision is jurisdictional and remedial; it supplies no substantive right of its own (42 U.S.C. 1983 2018). The plaintiff must therefore identify, for each defendant, the constitutional or statutory right infringed, the conduct that infringed it, and the causal link between the two. The five counts below are drafted to do exactly that.

3.2 Count I — Retaliatory arrest and prosecution

The first count pleads that the plaintiff was arrested and later prosecuted not on probable cause arising from criminal conduct but in retaliation for protected speech or petitioning. The doctrinal obstacle is well known: *Nieves v. Bartlett* holds that a plaintiff alleging a retaliatory arrest must plead and prove the absence of probable cause for the arrest (Nieves v. Bartlett 2019). The narrow exception the Court recognised — for jurisdictions in which otherwise similarly situated individuals are not arrested — has itself been narrowed: *Gonzalez v. Trevino* requires the plaintiff to identify objective evidence that the exception applies rather than to rely on the absence of comparators (Gonzalez v. Trevino 2024). A count pleaded without that evidence invites dismissal, and the pleading must therefore be built on specific, record-based comparators or on an identified policy of selective enforcement.

The draft complaint conditions this count on a showing that the arrest lacked probable cause and that the retaliatory motive was a but-for cause. The source portfolio itself flags that the underlying docket entries and the charging instrument must be verified before the count is filed [CITE REQUIRED].

3.3 Count II — Coercive interrogation and the privilege against self-incrimination

The second count pleads that statements were obtained under conditions that overbore the plaintiff's will, in violation of the Fifth and Fourteenth Amendments. The inquiry is whether the confession was the product of coercion rather than of a free choice. Because the count sounds in the conduct of interrogation rather than in the fact of prosecution, it must be analysed separately from Count I: the case law that limits retaliation claims does not automatically limit a coercion claim, and vice versa.

3.4 Count III — Fabrication of evidence and reliance on uncorroborated evidence

The third count pleads that the investigation relied on fabricated or uncorroborated material. Its doctrinal home is the line of authority that makes it actionable to fabricate evidence even where no conviction results, because the wrong is the creation of false evidence rather than the outcome of the case. The count must be pleaded against the specific officer who is said to have fabricated a specific thing, with dates and documents, and not against the investigating unit collectively (Buckley v. Fitzsimmons 1993).

3.5 Count IV — Conspiracy

The fourth count pleads agreement among the defendants. A conspiracy count under § 1983 must allege a meeting of the minds sufficient to support an inference of agreement, and it must identify the constitutional wrong that the agreement produced. It is not a vehicle for adding defendants whose only connection to the case is a shared employer.

3.6 Count V — Municipal and supervisory liability

The fifth count pleads that the constitutional injuries were caused by a policy or custom of the local government, or by a failure to train that amounts to deliberate indifference. *Monell* established that a municipality may be sued under § 1983 where the execution of its policy or custom inflicts the injury (Monell v. Department of Social Services 1978). The count must identify the policy, the practice, or the training deficiency, and must show that the deficiency was the moving force behind the violation. *City of Canton v. Harris* supplies the deliberate- indifference standard for training claims (City of Canton v. Harris 1989), and *Connick v. Thompson* cautions that a single act of omission does not ordinarily establish the required pattern (Connick v. Thompson 2011).

3.7 Relief and the structure of the prayer

The combined prayer seeks declaratory relief, injunctive relief, and damages, and each form of relief carries its own threshold. Injunctive relief requires the plaintiff to satisfy the four *Winter* factors by a clear showing: likelihood of success on the merits, likelihood of irreparable harm, a favourable balance of equities, and service of the public interest (Winter v. Natural Resources Defense Council 2008). Damages against individual defendants in their personal capacities require the plaintiff to overcome qualified immunity, the subject of Chapter 4. A prayer that omits the *Winter* showing asks the court to grant equitable relief it has no power to grant, and a prayer that omits the immunity analysis asks the court to skip the question it must answer first.

CHAPTER 4

Appellate Posture and Doctrinal Thresholds

Appellate jurisdiction is not an afterthought. It is a pleading requirement, because an order that cannot be appealed is an order the plaintiff has lost.

4.1 Finality and the collateral-order doctrine

The default rule is that only final decisions are appealable, under 28 U.S.C. § 1291 (28 U.S.C. 1291 2018). An order that contemplates further proceedings is not final, and the plaintiff who wishes to appeal it must find a foothold in the collateral-order doctrine of *Cohen v. Beneficial Industrial Loan Corp.*, which permits immediate appeal of orders that conclusively determine a disputed question, resolve an important issue separable from the merits, and would be effectively unreviewable on appeal from final judgment (Cohen v. Beneficial Industrial Loan Corp. 1949). The denial of a motion to dismiss on qualified-immunity grounds is the paradigm collateral order, and it is the principal route by which a civil-rights plaintiff's case reaches the circuit court before trial.

4.2 Qualified immunity

Qualified immunity shields government officials from civil damages liability unless their conduct violated a clearly established statutory or constitutional right of which a reasonable person would have known (Harlow v. Fitzgerald 1982). The doctrine has two prongs — whether a right was violated, and whether the right was clearly established at the time — and the Supreme Court has repeatedly permitted courts to resolve the second without deciding the first. For the plaintiff, the practical task is to identify a controlling precedent or a robust consensus of cases placing the unlawfulness of the conduct beyond debate.

4.3 Prosecutorial and absolute immunity

Prosecutors enjoy absolute immunity for conduct intimately associated with the judicial phase of the criminal process (Imbler v. Pachtman 1976), including administrative acts connected to the prosecution such as the handling of disclosure obligations (Van de Kamp v. Goldstein 2009). They enjoy only qualified immunity, however, for conduct that is investigative rather than advocative (Buckley v. Fitzsimmons 1993). The distinction matters because a complaint that seeks damages from a prosecutor must plead conduct on the correct side of the line.

4.4 Sovereign immunity and the *Ex parte Young* exception

The Eleventh Amendment generally bars suits for damages against a state in federal court, and states may not be sued as § 1983 defendants at all. *Ex parte Young* preserves a narrow exception: a suit for prospective injunctive relief against a state officer to end a continuing violation of federal law is not barred (Ex parte Young 1908). The remedy is against the officer, not the state, and it is prospective, not compensatory.

4.5 The favourable-termination element

Where a § 1983 claim implies the invalidity of a conviction or sentence, the plaintiff must show that the conviction or sentence has been reversed, expunged, or otherwise called into question. *Heck v. Humphrey* states the rule for malicious prosecution and related claims (Heck v. Humphrey 1994), and *Thompson v. Clark* clarifies what counts as favourable termination for a malicious-prosecution claim: a criminal defendant need not show that the prosecution ended in an affirmative indication of innocence, only that it ended without a conviction (Thompson v. Clark 2022). *Manuel v. City of Joliet* holds that a pretrial-detention claim is governed by the Fourth Amendment rather than by a fabricated a retrospective malice standard (Manuel v. City of Joliet 2017).

4.6 Abstention

Younger v. Harris directs federal courts to abstain from enjoining pending state criminal proceedings absent extraordinary circumstances (Younger v. Harris 1971). The doctrine is not absolute: bad faith, harassment, or a patently unconstitutional statute can lift it. A federal complaint that seeks to enjoin a state prosecution must therefore plead the bad-faith or harassment exception with specificity, or risk dismissal on abstention grounds.

CHAPTER 5

Extraordinary Writs and Supreme Court Practice

A petition for an extraordinary writ is a request to a court to reach material it would rather leave to the ordinary process. It is granted rarely, and the petition must therefore be the most disciplined document in the case.

5.1 The writs and their standards

The extraordinary writs available under the All Writs Act — mandamus, prohibition, and certiorari from the Supreme Court — differ in standard and in the office they perform. Mandamus compels a lower court or an officer to perform a clear legal duty; it is granted only where the petitioner has a clear and indisputable right, the respondent has a plain duty to act, and no other adequate means of relief exists. Prohibition restrains a court from acting outside its jurisdiction. A petition for a writ of certiorari asks the Supreme Court to exercise its discretionary review.

5.2 Drafting discipline for a petition

The source portfolio treats the petitions as working instruments, and the drafting discipline is worth stating explicitly. A petition must state the question presented in a single sentence; must identify the court that entered the judgment and the date of entry; must set out the facts with record citations; and must explain why the ordinary appellate process is inadequate. Every factual assertion that depends on a docket entry carries a citation marker until the entry is in hand.

Record proofs that the petitions require

The source materials identify, as necessary to any petition, the file-stamped pleadings, the docket sheets, and the orders actually entered in each proceeding **[CITE REQUIRED]**. Without them the petition asserts procedural history that the court cannot verify, and a court that cannot verify the procedural history will not reach the merits.

5.3 The Rule 60 and Rule 59 motions

Relief from judgment under Rule 60(b) and amendment or alteration of judgment under Rule 59(e) are frequently the prerequisites to appellate review of a procedural defect. A motion under either rule must be filed within the applicable time and must identify the specific ground. The source materials record the author's intention to preserve every available post-judgment motion, and the chapter treats that intention as a checklist rather than as a description of steps already taken.

CHAPTER 6

The Constitutional Case for Genomic Data Sovereignty

The genome is the most intimate instrument of prediction that has ever been readable. A Fourth Amendment built for papers and effects has to say something about it.

6.1 The Fourth Amendment and informational privacy

The Fourth Amendment protects persons, houses, papers, and effects against unreasonable searches. In the physical world it regulates the collection of things. In the digital world the Supreme Court has recognised that the collection of information can itself be a search, even when the information is held by a third party: *Carpenter v. United States* held that the government's acquisition of historical cell-site records was a search (Carpenter v. United States 2018), and *Riley v. California* held that a search incident to arrest cannot extend to the contents of a cell phone (Riley v. California 2014). *Maryland v. King* upheld DNA sampling of arrestees on the strength of the reduced expectation of privacy of persons in custody, but its reasoning is expressly limited to identification (Maryland v. King 2013). The analytical route from these cases to a general protection for genomic data is not automatic, and the argument for it proceeds by analogy: a genome is a record that is both identifying and predictive, and predictive capacity is a property the Court has treated as constitutionally significant.

6.2 The five properties of genomic information

The author's portfolio advances five properties that distinguish genomic information from other categories of data and that, taken together, justify a heightened standard.

1. **Uniqueness.** A genome is effectively unique to an individual and is stable across the lifespan.
2. **Relationality.** A genome is not the property of one person alone; it discloses probabilistic information about relatives who have not consented to anything.
3. **Predictive power.** A genome carries probabilistic information about future disease, and thus about future cost.
4. **Persistence.** A genome does not change and cannot be reissued, unlike a password or a credit-card number.

5. **Inter-generational reach.** The information is inherited and inherited-into, so the interest it creates outlives the individual.

6.3 The counter-move and its limit

The strongest counter-argument is that genomic information is neither more nor less private than other sensitive data, and that a special constitutional rule would be unmanageable. The limit of that argument is that three of the five properties — relationality, persistence, and inter-generational reach — are properties that no other category of personal data possesses to the same degree. A fourth, predictive power, is what the Court has treated as decisive in the third-party doctrine cases. The constitutional argument is not that the genome is unique, but that the combination of properties makes the existing, category-by- category approach inadequate.

CHAPTER 7

Statutory Architecture: GINA, the ADA, HIPAA, and Beyond

The genome is protected today by a lattice of statutes, each of which was drafted for a neighbouring problem. The gaps between them are the subject of this chapter.

7.1 GINA

The Genetic Information Nondiscrimination Act of 2008 prohibits discrimination on the basis of genetic information in health insurance and in employment (Genetic Information Nondiscrimination Act of 2008 2008). It is the only federal statute aimed directly at genetic discrimination, and its scope is the source of its principal gap. In the insurance market, GINA governs health insurance; it does not reach life insurance, disability insurance, or long-term-care insurance. In employment, it governs hiring, firing, and terms of employment, but it does not regulate an employer's use of genetic information in a wellness programme except as provided. The gap is structural: the statute regulates health underwriting precisely where genetic information is least powerful, and leaves unregulated the markets in which genetic information is most predictive.

7.2 The ADA and the amendments of 2008

The Americans with Disabilities Act prohibits discrimination against a qualified individual with a disability (Americans with Disabilities Act of 1990 1990), and the ADA Amendments Act of 2008 broadened the definition of disability so that the question would turn on whether discrimination occurred rather than on whether the plaintiff crossed a threshold (ADA Amendments Act of 2008 2008). For the genomic plaintiff the significance is twofold. First, a genetic predisposition can, in some circumstances, establish a record of impairment. Second, and more importantly for the disability context, *Olmstead v. L.C.* requires states to provide services to persons with disabilities in community settings rather than institutions when the state's treatment professionals determine that community placement is appropriate (Olmstead v. L.C. ex rel. Zimring 1999). A state agency that denies a benefit on the ground of a genetic prediction, without an individualised assessment, states a claim under the integration mandate.

7.3 HIPAA and the Security Rule

The Health Insurance Portability and Accountability Act and its implementing regulations establish national standards for the electronic exchange of health information and, under the Security Rule, for

the administrative, physical, and technical safeguards that protect it (Health Insurance Portability and Accountability Act of 1996 1996). The Security Rule expressly contemplates that encryption is not always feasible, and thus treats it as an addressable rather than a mandatory implementation specification. That flexibility, combined with the fact that most genomic data moves between systems that are not all covered by HIPAA, leaves a substantial middle ground in which genomic information is neither clearly protected nor clearly unprotected. The technical remedies developed in Parts IV and VI are aimed precisely at that middle ground.

7.4 Beyond the lattice

Two further instruments matter to the argument. The Americans with Disabilities Act's accessibility provisions, as elaborated in Section 504 of the Rehabilitation Act, supply the template for the cognitive-accessibility proposal developed in Part X. And the patchwork of state genetic-privacy statutes supplies examples of what a federal standard could look like. The chapter's conclusion is that the lattice is real but incomplete, and that the incompleteness is the space in which both the litigation theory and the technical proposal operate.

Part III

Neurogenetic Biomarkers and Algorithmic Governance

CHAPTER 8

Multimodal Biomarker Characterization

A prediction is not an observation, and a probability is not a diagnosis. The whole difficulty of neurogenetic evidence is that the two are easy to confuse in a document an adjudicator must read quickly.

8.1 The dissertation problem

The technical core of the author's scientific work is the characterisation of biomarkers for a neurodevelopmental phenotype using multiple modalities at once. The modalities fall into four families: behavioural and clinical instruments, functional neuroimaging, genotypic and polygenic measures, and physiological time-series such as the electroencephalogram. The dissertation problem is to characterise the phenotype as a function of these modalities jointly, and to report the uncertainty of the joint characterisation honestly.

8.2 Why multimodal beats unimodal, and when it does not

A multimodal characterisation can outperform a unimodal one when the modalities carry complementary information — that is, when the correlation between two modalities, conditional on the phenotype, is low. It does not outperform when the modalities are redundant, and it can underperform when one modality is noisy and its error is independent of the signal, because the fusion step then injects variance without adding information. The practical consequence is that a multimodal claim must be accompanied by a diagnosis of complementarity; a modality that merely restates another is not evidence, it is redundancy dressed as rigour.

8.3 The four families in detail

Clinical and behavioural instruments. Standardised instruments supply the reference phenotype. Their virtue is interpretability; their vice is subjectivity at the margins, and it is at the margins that adjudicative decisions are made.

Functional neuroimaging. Functional magnetic resonance imaging measures a blood-oxygenation-dependent signal that is a proxy for neural activity. Its strengths are spatial resolution and repeatability across scanners when a common protocol is used; its weaknesses are temporal resolution, cost, and sensitivity to motion and to analysis choices. The reproducibility literature on task-based fMRI is the reason this book

treats a single-study imaging result as evidence to be corroborated rather than as a stand-alone proof of anything.

Genotypic and polygenic measures. A polygenic score aggregates the small effects of many loci into a single number. Its predictive power depends on the discovery sample from which the weights were estimated, and it transfers least well across ancestries that were underrepresented in that sample. A polygenic score is therefore best understood as a population statistic applied to an individual, with an ancestry-dependent penalty that must be disclosed.

Physiological time-series. The electroencephalogram and related signals supply a temporal modality whose features are computable and whose analysis is comparatively transparent. They are the modality most amenable to the evidentiary pipeline described in Part VI, because the transformation from raw signal to feature is deterministic and auditable.

8.4 Uncertainty as a first-class output

The design decision that most distinguishes this work from a conventional biomarker study is that uncertainty is an output rather than a caveat. A characterisation that reports "score = 0.82" is less useful to an adjudicator than one that reports a posterior distribution, because the adjudicator's task is to decide under uncertainty rather than to rank point estimates. Chapter 10 returns to this point when it asks what a court should be allowed to do with such a distribution.

CHAPTER 9

Candidate SNP and Functional Variant Profiling

The unit of genetic evidence is not the gene. It is the variant, and the honest statement of what a variant does is a distribution over functions, not a verdict.

9.1 From genes to variants

A candidate-gene study nominates a gene and looks for association. A variant-level study nominates a base change and looks for association with a functional consequence. The second is the more defensible unit because it is the more specific one, and because the functional annotation that gives the association its meaning attaches to the variant rather than to the gene. The portfolio's profiling work proceeds at the variant level.

9.2 The classes of variant

The variants that matter for a neurodevelopmental phenotype fall into four classes. Coding variants alter an amino-acid sequence and admit a comparatively direct interpretation. Regulatory variants alter the expression of a gene and require a functional assay or a population-scale expression analysis to interpret. Copy-number variants alter the dosage of a region and require a platform capable of detecting structural change. And common variants of small effect contribute to a polygenic score. A profiling pipeline that treats these four classes identically will mis-attribute effect to whichever class its assay detects best.

9.3 The handheld electrochemical genotyping platform

The hardware element of this part is a portable genotyping platform built around an electrochemical readout and an addressable electrode array. The design's claim is that a sixteen-channel or sixty-four-channel array can perform allele-specific detection of a panel of variants in a point-of-care setting, trading breadth for locality. The engineering argument for the platform is set out in the source material's specification; its status here is **[DESIGN TARGET]**, because the performance figures are design estimates rather than measurements from a built unit.

An electrochemical readout transduces hybridisation into a current, which is easier to instrument in a portable form factor than fluorescence. The trade is sensitivity: fluorescence detects fewer copies, while electrochemistry favours ruggedness and cost. For a panel of common variants with high allele frequency, the trade is favourable; for rare variants or low-input samples, it is not, and the specification should say so.

9.4 The 96-electrode array and its addressability

A ninety-six-electrode array is chosen because ninety-six is the standard microtitre plate format and therefore composes cleanly with existing laboratory workflow. Its addressability is the feature that matters analytically: an individually addressable electrode allows the readout to be spatially resolved, so that a failure at one site is a datum rather than a corruption of the whole array. This is the same design principle that Part VI applies to the evidentiary pipeline.

9.5 What the platform does not claim

The platform does not claim to sequence a genome; it claims to interrogate a panel. It does not claim clinical diagnostic authority; it claims to supply a structured measurement that can be entered into a governed record. And it does not claim ancestry-independence; the panel is specified for a population and its transferability is a stated limitation.

CHAPTER 10

Daubert Admissibility of fMRI and Polygenic Scores

The question for the court is not whether the science is true. It is whether the witness's use of the science in this case is reliable enough to help the trier of fact.

10.1 The framework

Daubert v. Merrell Dow Pharmaceuticals requires the trial judge to act as gatekeeper for expert testimony, assessing whether the reasoning or methodology underlying the testimony is scientifically valid and whether it can be applied to the facts at issue (Daubert v. Merrell Dow Pharmaceuticals 1993). Rule 702 of the Federal Rules of Evidence codifies the inquiry and, as amended, requires that the testimony be based on sufficient facts or data, be the product of reliable principles and methods, and reflect a reliable application of those principles and methods to the facts (Federal Rule of Evidence 702 2023). *Kumho Tire v. Carmichael* extends the gatekeeping obligation to all expert testimony, not only to testimony that is scientific in the narrow sense (Kumho Tire Company v. Carmichael 1999).

10.2 fMRI in the courtroom

Functional imaging enters a case in one of two postures. In the first, it is offered as a diagnostic datum — a measurement that a clinician used in forming an opinion. In the second, it is offered as a direct index of a mental state, such as truthfulness or intent. The second posture has been received with scepticism, and rightly: the inferential distance from a blood-oxygenation-dependent signal to a proposition about intent is long, and it is not made shorter by the vividness of the image. A court confronted with an fMRI exhibit should ask three questions. What is the proposition the exhibit is offered to prove? What is the error rate of the inference from signal to proposition? And what would the exhibit change about the adjudicator's decision?

An image is more persuasive than a number, and a coloured brain map is more persuasive than a table. The gatekeeping inquiry is designed to interrupt that persuasiveness and to ask what the exhibit actually establishes. A brief that offers an imaging exhibit should therefore state the inference it supports in a single sentence before it shows the image.

10.3 Polygenic scores as evidence

A polygenic score is a weighted sum of allele counts. Its admissibility raises four methodological questions. First, *predictive validity*: what is the score's out-of-sample predictive performance for the phenotype at issue? Second, *transferability*: how does that performance degrade when the score is applied to an individual of different ancestry from the discovery sample? Third, *base rate*: what does a given score mean in the population at issue, as opposed to the study population? Fourth, *individualisation*: a score is a population statistic, and the step from a population distribution to a statement about a named individual is the step that most often fails.

The correct use of a polygenic score in adjudication is therefore as one input to a probabilistic assessment, with its confidence interval and its ancestry- dependent penalty disclosed. The incorrect use is as a categorical statement that an individual does or does not have a condition.

10.4 Evidentiary fusion and the record

The portfolio's solution to the admissibility problem is procedural rather than substantive. Every neurogenetic exhibit is routed through a pipeline that records the raw datum, the transformation applied, the intermediate results, and the model that produced the final number, so that a reviewing court can audit the inference. That pipeline is the OpenSSA integrity layer described in Part VI. The substantive point of this chapter is simply that an exhibit that cannot be audited is an exhibit that cannot be assessed for reliability under Rule 702, and an exhibit that cannot be assessed cannot be admitted.

Part IV

Hybrid Semantic Processing Architecture (HSPA v3.0)

CHAPTER 11

Architectural Philosophy and the Clean-Slate Mandate

Every instruction set carries a burden: the cost of translating what the programmer meant into what the silicon can do. HSPA's wager is that for some workloads the burden has become larger than the work.

11.1 The premise

HSPA — the Hybrid Semantic Processing Architecture, version 3.0 — is specified as a clean-slate processor design. It discards binary backward compatibility with x86, ARM, RISC-V, and SPARC, and executes natively on directed acyclic graphs of semantic nodes. The motivating diagnosis is the von Neumann bottleneck: the separation of memory from computation, combined with the sequential fetch-decode-execute cycle, which the specification characterises as a "semantic tax" because each instruction must be decoded and scheduled through layers of abstraction that discard context.

Register of the claims in this part

The design in this part is reported in the register of the source specification. Where a parameter is a published standard — for example the FIPS 203 key and ciphertext sizes — it is **[ADOPTED LAW]**. Where a parameter is a property of the author's simulation — for example the modelled nitrogen-vacancy coherence times — it is **[MODEL / SIMULATION]**. Where a figure describes hardware not yet fabricated at the stated scale, it is **[DESIGN TARGET]**.

11.2 The comparison the specification draws

The table is best read as a set of architectural commitments rather than as a performance comparison. It asserts, correctly, that the four legacy families share a structural assumption — that computation is a sequence — and that HSPA rejects the assumption. Whether rejecting it pays for a given workload is an empirical question the specification does not settle.

Table 11.1: Legacy instruction architectures and the HSPA clean-slate replacement.

Legacy	Stated limitation	HSPA replacement
x86 (CISC)	Variable-length decoding; implicit serial semantics	Native SEG parsing; node-at-a-time graph dispatch
ARM (RISC)	Fixed 32-bit encoding; load/store bottleneck	128-bit semantic-node packets; content-addressable operands
RISC-V	Register-register pipeline driven by a program counter	Graph-edge traversal without a program counter
Dataflow	Deterministic token matching without uncertainty	Bayesian confidence-weighted token firing
NPU / TPU	Fixed matrix-multiplication acceleration	Morphable reconfigurable logic cells

11.3 The six-layer stack

HSPA organises its functions into a strict six-layer physical and logical stack. The book follows the specification's own numbering, in which the quantum substrate is Layer 0 and the cognitive interface is Layer 5.

Layer 5 — Application and cognitive interface.
The HARMONI-X cognitive prosthetic, the OpenSSA integrity pipeline (OSI-P), retrieval-augmented generation with ReAct agent loops, a precision-medicine trial workflow, and an evidentiary fusion engine that draws on public court, biomedical, and scholarly corpora.

Layer 4 — Security and post-quantum trust.
Key encapsulation, digital signatures, hash-based anchoring, and a semantic evidence trace ledger.

Layer 3 — Reconfigurable compute fabric.
An array of reconfigurable logic cells with digital-analog hybrid arithmetic units, a self-optimising microcode layer, and per-cell micro-cachelets.

Layer 2 — Probabilistic control and memory.
A probabilistic microcontroller, a fixed-point inference matrix, and a unified memory lattice with hybrid semantic-physical addressing.

Layer 1 — Semantic execution graph.
A 128-bit fixed node format, locality-sensitive context hashing, and a graph-eligibility firing protocol.

Layer 0 — Quantum substrate.
A density-matrix nitrogen-vacancy quantum emulator on the carrier board.

CHAPTER 12

The Six-Layer Stack: Execution Graph, Control, and Compute

12.1 Layer 1 — the 128-bit semantic node

Programs are graphs of fixed-width nodes. The base node occupies 128 bits and encapsulates opcode, confidence, context hash, two operands, an output count, and flags:

Table 12.1: 128-bit native semantic node layout.

Bits	Field	Width and meaning
[127:120]	Opcode	8 bits; instruction selector
[119:112]	Confidence	8 bits; firing weight
[111:80]	Semantic tag	32-bit locality-sensitive context hash
[79:48]	Operand A	32-bit unified-memory address
[47:16]	Operand B	32-bit unified-memory address
[15:8]	Output count	8 bits
[7:0]	Flags	8 bits

In an extended 256-bit cache line the node carries additional execution metadata: a 16-bit expected latency, a 16-bit energy budget, a 32-bit vector of downstream node identifiers, a 64-bit uncertainty vector encoded as eight 8-bit probabilities over possible outcomes, and a 32-bit quantum handle that points to a register in the quantum substrate.

12.2 The semantic instruction set

The specification defines native 8-bit opcodes grouped into seven families.

Table 12.2: HSPA opcode families and their executing units.

Range	Group	Representative mnemonics	Unit
0x00–0x0F	Arithmetic	ADD, SUB, MUL, DIV	RLC arithmetic unit
0x10–0x1F	Logic	AND, OR, XOR, NOT	RLC logic gate
0x20–0x2F	Memory	LOAD, STORE, PREFETCH	UML interface
0x30–0x3F	Control	JUMP, BRANCH, CALL	PMCU scheduler
0x40–0x4F	HSPA extensions	QRAND, ENCRYPT, ATTEST	PQC / NV-centre unit
0x50–0x5F	AI / neural	MAC, ACTIVATE, POOL	RLC stochastic unit
0xE0–0xE6	Quantum native	QINIT, QGATE_SINGLE, QMEASURE	LoongDiamond substrate

12.3 The firing protocol

A node fires when five conditions hold simultaneously. All input edges must carry valid tokens; the node's normalised confidence must meet or exceed the threshold $P_{\min}$ (default 0.70); the semantic distance between the node tag and the active context of the target cell must fall below a bound δ; the node's energy budget must fit within the available energy for the epoch; and, if the node carries a quantum handle, the density matrix must be normalised to within a tolerance ϵ. The protocol is a conjunction rather than a disjunction, which matters: it means that a node with abundant data but low confidence does not fire, and a node with high confidence but no energy does not fire either.

12.4 Layer 2 — the probabilistic microcontroller and the memory lattice

The probabilistic microcontroller replaces the program counter, the branch predictor, and the superscalar scheduler with a fixed-point inference matrix in Q15 format. Scheduling is by an entropy-reduction score:

$$\text{Schedule}(n) = \alpha\, P_{\text{success}}(n) + \beta\, \text{confidence}(n) - \gamma\, \text{entropy}(n), \tag{12.1}$$

with default weights $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$. Beliefs are updated by Bayes' rule as telemetry arrives:

$$P(\text{success} \mid e) = \frac{P(e \mid \text{success})\, P(\text{success})}{P(e)}. \tag{12.2}$$

The negative entropy term in (12.1) is the architectural counterpart of the epistemic caution that Parts III and IX insist upon: the scheduler prefers executions that reduce uncertainty, not merely executions that are likely to succeed.

Memory is content-addressable through a 64-bit unified address.

Table 12.3: 64-bit unified memory lattice address structure.

Bits	Field	Function
[63:48]	Physical page index	DRAM row and bank mapping
[47:16]	Semantic hash	Locality-sensitive hash of the content
[15:8]	Privilege and access	Read/write/execute plus quantum-enclave access
[7:0]	Uncertainty tolerance	Permitted error bound for approximate computing

Semantic distance between two addresses is computed in hardware as the Hamming distance of their semantic hashes:

$$d_{\text{sem}}(a, b) = \text{popcount}(\text{hash}_a \oplus \text{hash}_b). \tag{12.3}$$

The design makes semantic access primary and physical access secondary, which is the inversion that distinguishes it from a conventional cache.

12.5 Layer 3 — reconfigurable logic cells and self-optimising microcode

Each reconfigurable logic cell morphs in under five nanoseconds, carries a digital-analog hybrid arithmetic unit supporting precisions from one to thirty-two bits, a 1024-bit lookup table, and a two-kilobyte least-recently-used micro-cachelet. The specification reports an 85% hit rate on repeated semantic tags, a figure that is **[MODEL / SIMULATION]** and depends on the tag distribution.

The self-optimising microcode layer is an on-chip reinforcement-learning agent that rewrites microcode tables using proximal policy optimisation, with reward

$$R(s, a) = w_1 \cdot \text{Throughput} + w_2 \cdot \text{Energy Efficiency} + w_3 \cdot \text{Accuracy} - w_4 \cdot \text{Quantum Decoherence}. \tag{12.4}$$

The subtraction in (12.4) is the design's acknowledgement that coherence is a resource to be conserved. Safety is enforced by a commit protocol: ten thousand validation vectors, a block-RAM rollback buffer, a

signature check under a post-quantum scheme, and canary deployment across one per cent of cells over a thousand epochs. The protocol is the reason the layer can be described as self-optimising without being described as unsupervised.

CHAPTER 13

The LoongDiamond-2.5 Quantum Substrate

A density matrix is not a wavefunction. It is what a wavefunction looks like to an observer who has already given up on knowing everything.

13.1 The physical model

The substrate, named LoongDiamond-2.5, simulates nitrogen-vacancy (NV) centre qubits in synthetic diamond using a density matrix as the single source of truth. The density matrix ρ is Hermitian, of unit trace, and positive semi-definite, so $\rho = \rho^\dagger$, $\operatorname{Tr}\rho = 1$, and $\operatorname{Tr}\rho^2 \le 1$. The simulation's parameters are drawn from the NV-centre literature.

Table 13.1: LoongDiamond-2.5 NV-centre parameters as specified.

Parameter	Symbol	Value	Context
Temperature	T	4.0 K	Cryogenic operation
Magnetic bias field	B	500 G	Bias field
Zero-field splitting	D_{gs}	2.87 GHz	Ground-state D parameter
Spin-lattice relaxation	T_1	10.0 ms	NV baseline
Spin-spin coherence (echo)	T_2	5.0 ms	Echo-decoupled
Inhomogeneous dephasing	T_2^*	0.1 ms	Free-induction decay
Single-qubit gate time	t_{gate}	100 ps	Pulse duration
Single-qubit fidelity	F_{1q}	99.5%	Microwave drive
Two-qubit fidelity	F_{2q}	99.0%	Dipolar coupling
Measurement fidelity	F_{meas}	98.5%	Optical readout
Initialisation fidelity	F_{init}	99.9%	Optical spin pumping
Optical cycle time	t_{opt}	300 ns	Fluorescence cycle

The zero-field splitting of 2.87 GHz and the millisecond-order relaxation and coherence times are consistent with the experimental NV literature; the free-induction dephasing figure is consistent with the microsecond-scale values reported for NV ensembles in isotopically enriched diamond, where median T_2^* of several microseconds has been measured (Shinei et al. 2025). The table therefore separates cleanly into real constants and design targets, and the chapter maintains that separation.

13.2 Decoherence as a Kraus map

Decoherence is modelled with completely positive, trace-preserving Kraus operators, $\rho \mapsto \sum_i K_i \rho K_i^\dagger$. Amplitude damping models energy relaxation:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}, \quad \gamma = 1 - e^{-t/T_1}, \tag{13.1}$$

and phase damping models loss of coherence:

$$K_0 = \sqrt{1-\lambda}\, I, \quad K_1 = \sqrt{\lambda}\, Z, \quad \lambda = 1 - e^{-t/T_2}. \tag{13.2}$$

Choosing the Kraus formalism rather than a wavefunction collapse postulates is a deliberate modelling decision: it makes the substrate's state a probabilistic object throughout, which is what allows the scheduler of Chapter 12 to treat coherence as a schedulable quantity.

13.3 VQE validation

The substrate is validated against the ground-state energy of molecular hydrogen using a hardware-efficient RY-CNOT ansatz with the qubit Hamiltonian

$$H = -1.0523\,(Z_0 I_1 + I_0 Z_1) + 0.3979\, Z_0 Z_1 + 0.0113\, X_0 X_1. \tag{13.3}$$

The variational eigensolver converges to -1.1247 Ha against the theoretical -1.1373 Ha, an error of 1.1%, within chemical accuracy, in 47 iterations. The variational eigensolver is the appropriate validation target for a noisy device because it requires only constant-depth circuits and tolerates repeated measurement (Zhuang et al. 2024); the result is **[MODEL / SIMULATION]**, being a simulation of a simulation.

CHAPTER 14

Production Hardware: The ZU9EG Carrier and the SSE-9 Processor

14.1 The stochastic semantic engine

The production platform, SSE-9, implements a stochastic refinement scheme with an asymmetric duty cycle in which thirty-two latent registers are updated at twice the clock frequency and sixteen prediction registers at the base frequency. A hardware convergence trigger monitors the gradient norm of the latent state:

$$\text{Trigger} = \begin{cases} \text{active} & \text{if } \left\| \dfrac{z_{t+1} - z_t}{\Delta t} \right\|_2 < \varepsilon, \\ \text{idle} & \text{otherwise.} \end{cases} \tag{14.1}$$

When the trigger asserts, control of the register file passes deterministically to the projection phase, which produces the adjudication vector through a hyperbolic tangent. The asymmetry is the architecture's central efficiency claim: refinement runs hot, projection runs cold, and the trigger decides when to switch.

14.2 The carrier board

The board targets a Xilinx Zynq UltraScale+ MPSoC (XCZU9EG) paired with sixteen gigabytes of DDR4-2400 in a fly-by topology. The stackup is a ten-layer impedance-controlled design.

Table 14.1: Ten-layer impedance-controlled PCB stackup.

Layer	Type	Copper	Function
L1 (top)	Signal	0.5 oz + plating	High-speed signals, BGA pads, cell array
L2	Ground	1.0 oz	Solid reference plane
L3	Signal	0.5 oz	High-speed signal, DDR4 data, 100 Ω differential
L4	Power	1.0 oz	Core supply (0.85/1.0 V at 20 A)
L5	Signal	0.5 oz	Graph node bus, 85 Ω differential pairs
L6	Ground	1.0 oz	Core reference plane
L7	Signal	0.5 oz	Quantum emulator interconnect
L8	Power	1.0 oz	Auxiliary and DRAM supplies
L9	Ground	1.0 oz	Serial and PCIe return plane
L10 (bottom)	Signal	0.5 oz + plating	Cryptographic accelerator, passives

The stackup, the itemised bill of materials, the schematics, and the reference implementations together constitute the handoff package described in Chapter 16. Their status is **[DESIGN TARGET]**: they specify a board, they do not report a board in production.

CHAPTER 15

Post-Quantum Cryptography and RIGOR-PRS-Secure++

The migration to post-quantum cryptography is unusual among security projects in that the deadline is set by an adversary's capabilities rather than by one's own roadmap.

15.1 Naming: from competition entries to standards

The specification refers to algorithms by their round-three competition names — "Kyber" and "Dilithium". Because those names have been superseded by the standards that adopted them, this book uses the standardised names throughout and records the mapping. Kyber-768 became ML-KEM-768 and Kyber-1024 became ML-KEM-1024 in FIPS 203; Dilithium-III became ML-DSA-65 and Dilithium-V became ML-DSA-87 in FIPS 204. The National Institute of Standards and Technology published FIPS 203, FIPS 204, and FIPS 205 in August 2024, and the National Security Agency's Commercial National Security Algorithm Suite 2.0 sets a transition timeline that reaches full adoption in the mid-2030s (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d; National Security Agency 2022a).

Table 15.1: Standards and algorithm mapping for the trust chain.

Standard	Body	Primitive	Sizes
FIPS 203	NIST	ML-KEM-1024	pk 1568 B, ct 1568 B, ss 32 B
FIPS 204	NIST	ML-DSA-87	pk 2592 B, sk 4896 B, sig 4627 B
FIPS 205	NIST	SLH-DSA-SHA2-128s	Stateless hash-based signature
SP 800-208	NIST	HSS-LMS	Stateful two-level LM tree

15.2 The dual-envelope construction

The trust chain wraps a payload in a layered envelope it calls QRDE-AEAD. A recipient key encapsulation yields a ciphertext and a shared secret,

$$\mathrm{ML\text{-}KEM.Encaps}(pk_{\mathrm{recv}}) \to (c_1, ss), \tag{15.1}$$

from which a master key is derived,

$$K_{\mathrm{master}} = \mathrm{HKDF\text{-}SHA3\text{-}384}(ss, \text{“QRDE-Master-Salt”}), \tag{15.2}$$

and split into a confidentiality key and a MAC key, $K_e = K_{\mathrm{master}}[0..31]$ and $K_m = K_{\mathrm{master}}[32..63]$. The plaintext is encrypted under an authenticated cipher,

$$ct_2, \tau_2 = \mathrm{AES\text{-}256\text{-}GCM.Encrypt}(K_e, IV, P, AD_{\mathrm{inner}}), \tag{15.3}$$

and the outer envelope is authenticated to bind the encapsulation to the ciphertext,

$$\tau_1 = \mathrm{HMAC\text{-}SHA3\text{-}256}(K_m,\ c_1 \,\|\, ct_2 \,\|\, \tau_2 \,\|\, AD_{\mathrm{outer}}). \tag{15.4}$$

The envelope $C = (c_1, IV, \tau_2, ct_2, \tau_1, AD_{\mathrm{outer}})$ is specified to achieve indistinguishability under adaptive chosen-ciphertext attack against a quantum adversary. The construction is a standard hybrid KEM-plus-DEM composition, and its security follows from the security of the components under the standard composition theorem; the contribution here is the particular binding of context through the two associated-data strings.

15.3 Where the security argument is strongest and weakest

The strongest part of the argument is the KEM: ML-KEM-1024 targets the highest of the NIST security categories and its parameters are fixed by a published standard. The weakest part is the operational one. A hybrid envelope is only as strong as the key management beneath it, and the source material’s own requirement that keys be derived, split, and bound to context is precisely the requirement that is most often omitted when such a scheme is deployed. The chapter therefore states the cryptographic claim and immediately states its dependency: the envelope protects a payload if and only if the recipient’s private key is protected by equal care.

CHAPTER 16

Cognitive Interface, Exhibits, Patent Claims, and Verification

16.1 Layer 5 — the cognitive interface

The application layer contains the HARMONI-X cognitive prosthetic, which the specification describes as an accessibility interface that adapts presentation to measured cognitive load. Its features include adaptive load monitoring drawn from heart-rate variability, pupilometry, and keystroke dynamics; display safeguards including pulse-width-modulation-free output, a dyslexia-friendly typeface, and a 21:1 contrast ratio; and a resource-allocation mode that can devote four times the nominal capacity to a task when a neurogenetic metabolic tax is present. The interface is the part of the architecture with the clearest legal hook, because it is the part that implements an accommodation, and Part X develops that hook.

The layer also contains the OpenSSA integrity pipeline, described in Part VI, and an evidentiary fusion engine that assembles material from public court dockets, biomedical repositories, and scholarly indexes.

16.2 The 16-marker crosswalk

The specification includes a crosswalk that maps sixteen genetic markers to functional impacts and to the listings used in disability adjudication.

Table 16.1: Selected markers from the 16-marker SNP-to-listing crosswalk.

rsID	Gene	Functional impact	Listing
rs4680	COMT	Slow dopamine clearance (Val158Met)	11.00 / 12.02
rs1800497	DRD2/ANKK1	Reduced D2 density (Taq1A)	12.11
rs2075654	SLC6A3	DAT1 transporter variability	11.00B / 12.02A
rs1360780	FKBP5	HPA-axis dysregulation	12.15
rs6265	BDNF	Reduced neuroplasticity (Val66Met)	11.00F
rs1801133	MTHFR	Folate defect (C677T)	11.00 / 14.08
rs3131972	HLA-DQB1	Immunogenetic risk	14.08H
rs4475691	TPH2	Serotonin synthesis, mood	12.04

The crosswalk is a proposal about how genetic evidence could be indexed to adjudicative criteria. It is not a diagnostic instrument, and the association between any single marker and any listed impairment is far weaker than the table's structure suggests. Chapter 10's discussion of polygenic scores is the controlling caution.

16.3 System exhibits

The specification includes four system exhibits: the execution and scheduling graph; the post-quantum trust chain; the asymmetric convergence loop; and a forum matrix that maps the litigation strategy across the Court of Federal Claims, the Federal Circuit, the district court, the state court, and the bankruptcy court. The forum matrix is reproduced in Part IX, where it is treated as a description of intended practice rather than of filed events.

16.4 The patent claims

The clean-slate disclosure advances an independent claim and six dependents. The independent claim recites a processor architecture comprising a semantic instruction set defining nodes in a directed acyclic graph, each node carrying a 128-bit metadata word; a density-matrix quantum emulation substrate coupled to the graph; a probabilistic microcontroller unit scheduling node execution by Bayesian inference; reconfigurable logic cells with nanosecond morphing; a unified memory lattice in which semantic access is primary; a self-optimising microcode layer; and a post-quantum trust chain binding each node. The dependents add the hybrid quantum-classical substrate, the entropy-reduction scheduler, the morphable fabric, the proximal-policy microcode layer, the cognitive accessibility interface, and the native semantic memory format.

A claim drawn to a processor architecture is directed to a machine or manufacture and is not, on its face, excluded. The exposure is that a claim which recites a desired result — "scheduling by Bayesian inference" or "self-optimising microcode" — rather than a specific technical means may be rejected as claiming an abstract idea under *Alice Corp. v. CLS Bank International* (Alice Corporation v. CLS Bank International 2014). The dependent claims, which recite specific bit layouts, specific operator forms, and specific timing bounds, are the parts of the disclosure most likely to survive, precisely because they are the most concrete. This is **[PROPOSAL]** as to strategy and **[ADOPTED LAW]** as to the governing standard.

16.5 Verification and the handoff package

The specification closes with a verification plan and a list of deliverables: register-transfer-level scaffolds, reference implementations, printed-circuit designs, bills of materials, cryptographic constructions, and the crosswalk. The verification plan is the part that makes the rest auditable, and it is the part this book treats as the model for how the whole portfolio should present technical claims — a claim is a claim only when a test that could falsify it accompanies it.

Part V

The MORTAR Algorithmic Queueing Engine

CHAPTER 17

The MORTAR Triage Engine: Problem, Architecture, and Invariants

An administrative queue is a moral instrument. The order in which a state adjudicates claims determines who is served and who waits, and waiting is not free.

17.1 The problem: a queue that kills

The Social Security Administration adjudicates disability claims on a first-in, first-out basis, subject to statutory and regulatory priorities. First-in, first-out is fair in the sense that it treats arrival order as the only relevant variable, and it is efficient in the sense that it requires no information beyond a timestamp. It is defective in one respect that matters: it is indifferent to how much a claimant's welfare depends on the speed of the decision. When the population in the queue has heterogeneous mortality risk — and the disability population does, being by definition a population with serious medical impairment — an order-blind queue imposes the largest cost on the claimants least able to bear it.

The magnitude of the problem is documented. The Government Accountability Office has reported that over a decade, tens of thousands of claimants died while their appeals were pending, that median appeal processing times exceeded a year and in some years approached two, and that a substantial number of claimants filed for bankruptcy while waiting (U.S. Government Accountability Office 2020). The author's own figure of more than thirty thousand preventable deaths annually in the United States is consistent with the range reported in the literature and in congressional oversight, and is treated here as an order-of-magnitude claim rather than as a precise measurement (U.S. Government Accountability Office 2020; Sanders and Larson 2020).

Register

The existence of the backlog and of mortality during the wait is **[ADOPTED LAW]** in the sense that it is documented by official audit. The attribution of any specific death to the delay is a counterfactual claim that is inherently contestable and is not established by the aggregate figures. MORTAR is **[PROPOSAL]**; its performance is **[MODEL / SIMULATION]**.

17.2 The architecture in outline

MORTAR — Mortality-Optimized Risk-Triage with Adaptive Rescheduling — is a layered, time-indexed directed acyclic graph for administrative claim prioritisation. It organises computation into five layers,

ordered strictly by dependency, and it is designed to satisfy a set of invariants that the author states explicitly.

L0 — Ingestion and validation.
Each claim is represented as a tuple $c_i = (\mathrm{ID}_i, \mathrm{arrival}_i, \mathrm{stage}_i, \mathrm{evidence}_i, \mathrm{status}_i, \mathrm{privacy}_i)$. A claim that fails validation is never deleted; it is routed to a defined exception node — a request for missing evidence, a human review for an identity conflict, a merge review for a duplicate, or a recovery queue for corruption.

L1 — Feature extraction.
Mortality risk, vulnerability, and urgency features are computed from the claim record.

L2 — Priority construction.
A composite priority score is computed with an explicit anti-starvation term.

L3 — Bipartite resource matching.
Claims are assigned to adjudicators and resources under capacity constraints by a matching algorithm.

L4 — State transition.
Each state update is committed to the audit chain described in Part VI.

17.3 The invariants

The specification states nine invariants, of which the first is fundamental.

Principle 17.1 (Claim-state invariant)**.** *Every claim state possesses a predecessor, a processing rule, a resulting state, and an auditable transition record.*

The remaining invariants assert that every input is traceable to a processing node; every node has defined prerequisites; every priority calculation is reproducible; every allocation decision is auditable; every state transition produces a verifiable record; every exception has a defined transition; no claim can disappear silently from the queue; human review is available at every materially adverse transition; and the three petition branches remain logically distinct.

17.4 Priority construction and the anti-starvation guarantee

Layer L2 computes a composite priority score. The specification's form is a weighted sum of risk, vulnerability, and urgency features, to which an ageing term is added:

$$\mathrm{priority}_i(t) = w_r\,\mathrm{risk}_i + w_v\,\mathrm{vuln}_i + w_u\,\mathrm{urgency}_i + \lambda\,(t - \mathrm{arrival}_i), \tag{17.1}$$

with $\lambda > 0$. The last term is the anti-starvation device: because the priority of every waiting claim increases at rate λ, a claim that is repeatedly passed over eventually outranks any claim whose feature score is bounded. If feature scores lie in $[0, S]$ and the weights sum to one, then no claim can be starved beyond a waiting time of order S/λ, which gives the design its hard anti-starvation guarantee. The guarantee is the ethical core of MORTAR: a triage rule that reorders by risk and omits (17.1)'s ageing term is a rule that can postpone a low-risk claimant indefinitely, and indefinite postponement is the injury the engine exists to prevent.

17.5 Resource matching

Layer L3 assigns claims to adjudicators and to constrained resources by a capacity-aware bipartite matching, in which each claim is a node on one side, each adjudicator-and-slot is a node on the other, and the objective is to maximise total priority subject to capacity. The assignment is a min-cost matching problem and is

solved accordingly; the design's contribution is not the algorithm, which is standard, but the requirement that the assignment be committed to the audit chain so that the reason a claim waits is recoverable.

The invariants are the more important half of the design. A triage system that reorders a queue is a system that can harm the people it moves backward; the invariants are what prevent the reordering from becoming an invisible reallocation.

17.6 Jurisdictional decomposition

MORTAR is shared by three petition branches — domestic administrative (SSA), a United Nations human-rights branch, and a neurotechnology-rights branch — which share an evidentiary and algorithmic substrate while remaining logically distinct. The master graph is

$$v_0 \longrightarrow \{v_1, v_2, v_3\} \longrightarrow v_4 \longrightarrow \{v_5, v_6, v_7\},$$

with root the claimant's evidence, the three jurisdictional branches feeding a common algorithmic core, and the core producing evidence, remedy, and enforcement outputs that resolve to a verified, audited state. Separating the branches matters because the legal standards they apply differ: a domestic adjudication and an international petition are not the same proceeding, and a design that fused them would produce a record that satisfies neither forum.

CHAPTER 18

Locality-Sensitive Hashing: Theory, Simulation, and Fairness

A hash function that preserves similarity is a way of making the question "is this like that?" cheap. Cheap similarity is powerful, and power requires a fairness audit.

18.1 Why locality-sensitive hashing

The computational bottleneck in triage is comparison: deciding which of many claims are alike in the features that matter, without comparing every pair. Locality-sensitive hashing solves exactly this problem. A family of hash functions is locality-sensitive if similar inputs collide with higher probability than dissimilar ones.

Principle 18.1 (Locality-sensitive family)**.** *A family $\mathcal{H}$ is (r_1, r_2, p_1, p_2)-sensitive for a similarity measure if, for $p_1 > p_2$, any pair with distance at most r_1 collides with probability at least p_1 and any pair with distance at least r_2 collides with probability at most p_2.*

The construction the specification names is drawn from the p-stable hashing literature (Datar et al. 2004) and from the min-wise and random-projection schemes that preceded it (Gionis et al. 1999a). For a random projection with a p-stable distribution, the hash is

$$h_{a,b}(x) = \left\lfloor \frac{a \cdot x + b}{w} \right\rfloor, \tag{18.1}$$

and the collision probability for two points at distance d is

$$P\big[h(a) = h(b)\big] = \int_0^w \frac{1}{d}\, f_p\!\left(\frac{t}{d}\right)\left(1 - \frac{t}{w}\right) dt, \tag{18.2}$$

where f_p is the density of the stable distribution. For the angular random-projection variant, the collision probability has the closed form

$$P\big[h(a) = h(b)\big] = 1 - \frac{\theta(a,b)}{\pi}, \tag{18.3}$$

with θ the angle between the vectors. Equation (18.3) is the reason the design uses random projections: the collision probability is a monotone function of similarity, so tuning the hash tunes the triage threshold analytically rather than by trial.

18.2 Amplification

A single hash is too weak. Two standard amplifications are composed. The *and*-construction places k hashes in a band and requires all to agree, multiplying specificity and raising the false-positive resistance to $1 - p_1^k$. The *or*-construction uses L independent tables and accepts a match if any table agrees, raising recall to $1 - (1 - p_1^k)^L$. The design's parameters are the number of tables and the band size, and they are tuned to a target recall and precision pair.

Increasing k raises precision and lowers recall; increasing L raises recall and costs memory and query time. The pair (k, L) is therefore the triage engine's policy surface, and it should be reported as such — an engine that does not disclose its (k, L) does not disclose how many similar claims it will miss.

18.3 Complexity

For n claims, L tables, and band size k, the query cost is $O(L \cdot (k + C))$ where C is the number of candidates retrieved, against $O(n)$ for a linear scan. The design's claim is that the constant factor and the candidate count are small enough at the relevant n that the engine is practically cheaper than a scan; that claim is **[MODEL / SIMULATION]** and is falsifiable by benchmark.

18.4 Simulation and its limits

The specification reports a Monte Carlo simulation of the triage engine. Three limits should be borne in mind. First, a simulation of a priority rule is a simulation of the rule, not of the agency: it cannot model the political economy that shapes the queue. Second, the simulation's feature distribution is assumed rather than observed, and the result is therefore conditional on that assumption. Third, a simulated reduction in waiting time is not a reduction in mortality unless a hazards model links the two, and the hazards model is where the causal claim lives. The chapter's conclusion is that the simulation supports the engineering claim — the engine computes what it says it computes — and does not by itself support the policy claim.

18.5 Fairness audit

A triage engine that prioritises by predicted mortality risk will, by construction, prioritise the sick. If the mortality model is less accurate for a group that has been historically under-treated, the engine can deprioritise that group not because its members are healthier but because the model sees them less clearly. The mitigation specified in the design is twofold: the anti-starvation term of Chapter 17 guarantees that no claim waits beyond a bounded time, and the audit chain of Part VI records the features and the model version that produced each priority, so that a disparity can be discovered after the fact. Neither measure eliminates the disparity; both make it visible, and visibility is the condition of correction.

Part VI

The OpenSSA Integrity Pipeline and Post-Quantum Security

CHAPTER 19

The OpenSSA Integrity Pipeline

Evidence that cannot be reproduced cannot be rebutted, and evidence that cannot be rebutted is not evidence; it is assertion.

19.1 The problem of the untraceable record

An adjudicative record assembled by software has a property that a paper record does not: its intermediate states are ephemeral. A model ingests a document, transforms it, and produces a conclusion, and the transformation leaves no trace unless the system is built to leave one. The consequence is a record that can be authenticated at its endpoints — the input existed and the output exists — while being unfalsifiable in the middle. The OpenSSA integrity pipeline (OSI-P) is the author's answer: a nine-stage audit trail in which each stage commits a hash of its output to a chain that a later stage can verify.

19.2 The nine stages

The pipeline names nine stages, each of which corresponds to a step at which a generative or retrievally system can silently alter material.

1. **Source integrity.** The provenance and hash of each source document.
2. **Query integrity.** The exact query issued to the retrieval system.
3. **Retrieval integrity.** The identity of each retrieved item.
4. **Augmentation integrity.** The context assembled for the model.
5. **Generation integrity.** The model, version, and parameters that produced the output.
6. **Validation integrity.** The checks applied to the output and their results.
7. **Chain of custody.** The ordered handoff of the material between stages.
8. **Reversibility.** The ability to reconstruct intermediate states.
9. **Audit summary.** A single signed digest of the whole.

The stages are not independent checks; they compose. Reversibility is what makes the earlier stages worth recording, because a party who can recompute an intermediate state can test the claim that the state was what the record says it was.

19.3 The Merkle chain

Each stage's output is hashed with SHA3-256 and the digest is committed to a Merkle tree over the pipeline's outputs. The Merkle structure gives the property that a single root digest authenticates the whole transcript, while allowing any individual leaf to be proved included with a logarithmic-size proof. The design signs the root with a post-quantum signature (ML-DSA-87, the standardised name for the scheme the specification calls Dilithium-III/ML-DSA-87), so that the chain's integrity does not depend on the hardness of factoring or of discrete logarithms.

SHA3 is chosen because it is a standardised sponge construction with no known structural weakness of the kind that has mattered for older hashes. The Merkle tree is chosen because it converts a linear transcript into a logarithmic inclusion proof, which is what makes the audit usable in a court: a party can prove that a specific document was part of the record without producing the whole record. The two choices are independent, and each is defensible on its own.

19.4 Self-authentication under the evidence rules

The pipeline is designed so that its output can be admitted as a self- authenticating record under Rules 902(13) and 902(14) of the Federal Rules of Evidence. Rule 902(13) permits authentication of a record generated by an electronic process on the basis of a certification by a qualified person who describes the process; Rule 902(14) permits authentication of a copy of data by a hash value and a certification. The pipeline's design maps onto both provisions directly: the Merkle root is the hash value of Rule 902(14), and the stage certifications are the descriptions of process required by Rule 902(13).

Rule 902(13) and 902(14) authenticate a record; they do not establish that the record is true. A chain that proves the model was version 4.2 in June does not prove that version 4.2 was accurate, fair, or lawful. The pipeline is a condition of admissibility, not a substitute for the substantive inquiry, and a brief that treats a clean audit chain as proof of correctness has confused the two.

19.5 The evidentiary fusion engine

The application layer assembles material from public court dockets, biomedical repositories, and scholarly indexes, and routes it through the pipeline. This is the point at which the architecture touches the litigation theory of Part II: the route by which an exhibit becomes admissible is the route the pipeline records, and the record it produces is the record the *Daubert* inquiry of Chapter 10 requires.

CHAPTER 20

The Post-Quantum Security Module

The data a state collects about its citizens has a long tail of sensitivity. A cipher whose security expires before the data does is a cipher that has already failed, quietly.

20.1 The threat model

Two quantum threats bear on data infrastructure. The first is the retrospective one: an adversary records encrypted traffic today and decrypts it when a sufficiently capable quantum computer exists — the "harvest now, decrypt later" pattern. Because genomic, medical, and legal records are sensitive for decades, a record encrypted under a scheme that a quantum computer can break is effectively already exposed. The second is the prospective one: an adversary forges a digital signature, undermining the authenticity of records rather than their confidentiality.

The National Security Agency's Commercial National Security Algorithm Suite 2.0 sets a transition timeline with software and firmware signing moving first and full adoption following (National Security Agency 2022a). The NIST standards published in August 2024 supply the primitives (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d).

20.2 The module's components

The security module comprises four components: a key-management service that derives and escrows post-quantum keys under policy; the dual-envelope authenticated-encryption construction of Chapter 15; a signature service for records and for the audit chain; and a zero-knowledge proving service that answers eligibility and priority queries without disclosing the underlying record.

20.3 Zero-knowledge proofs in the adjudicative context

A zero-knowledge proof allows a prover to demonstrate that a statement is true without revealing why it is true. The adjudicative use is precise: a claimant can prove that a priority threshold is met — for example, that a computed risk feature exceeds a value — without disclosing the genetic or clinical data from which the feature was computed. The property is valuable because the alternative is a system in which the claimant must expose the most sensitive data available about herself in order to be moved up a queue.

Status of the zero-knowledge component

That zero-knowledge proof systems exist and can prove statements of the relevant form is **[ADOPTED LAW]**. That the specific circuit needed here has been built, and that its proof cost is acceptable at the scale of a national adjudication queue, is **[DESIGN TARGET]**. The chapter states the design and names the gap.

20.4 Key management is the whole of the matter

The module's strongest cryptographic component is also its least novel, and its weakest component is the one that is easiest to neglect. A post-quantum KEM protects a payload only if the recipient's private key is protected; a post-quantum signature authenticates a record only if the signing key is not compromised. The history of cryptographic failure is overwhelmingly a history of key management failure, and the module's specification of key derivation, split, and binding is the part of Part VI that deserves the most scrutiny in deployment.

Part VII

International Human Rights Framework

CHAPTER 21

Request for a Commission of Inquiry

Domestic remedies fail silently. International mechanisms exist precisely for the case in which the ordinary process has become the instrument of the harm.

21.1 The instrument and its jurisdictional basis

The petition asks the United Nations Human Rights Council to establish a commission of inquiry into the administration of disability adjudication in the United States, and to refer questions to the International Court of Justice for an advisory opinion. The procedural vehicle is the Human Rights Council's complaint procedure, and the substantive basis is a set of treaty obligations the United States has assumed: the International Covenant on Civil and Political Rights, the Convention against Torture, and — as a matter of the norms invoked even where formal ratification is absent — the Convention on the Rights of Persons with Disabilities. The petition invokes the jus cogens character of the prohibition on arbitrary deprivation of life.

Register and a necessary caution

The petition is **[PROPOSAL]**: it is a request, not a filed instrument, and the establishment of a commission of inquiry is a discretionary act of the Council. The treaty architecture it invokes is **[ADOPTED LAW]**. Where the petition characterises aggregate mortality as a violation, the characterisation is an argument about attribution, and it is contested; the book preserves the distinction between the documented facts and the legal conclusion drawn from them.

21.2 The factual predicate and its sources

The petition rests its factual predicate on four categories of official and scholarly material. The first is the mortality of claimants awaiting adjudication: the Government Accountability Office reported a documented figure on the order of one hundred nine thousand seven hundred twenty-five appellants who died between 2008 and 2019 while their appeals were pending (U.S. Government Accountability Office 2020). The second is the scale of the backlog and its duration. The third is the reported association between benefit removal and subsequent incarceration, for which the petition cites the empirical literature on the criminal-justice consequences of disability-benefit loss. The fourth is the reported elasticity of mortality with respect to disability income, cited to the peer-reviewed literature (Gelber et al. 2023).

The table is the petition's strongest asset and, read carefully, its most fragile. The first row is a count from an official audit. The later rows are estimates with confidence intervals and modelling assumptions,

Table 21.1: The petition's quantified factual predicate.

Metric	Value	Source
Documented deaths, appellants	109,725	GAO-20-641R (U.S. Government Accountability Office 2020)
Deaths, pending claims FY2017–2023	133,000+	Congressional oversight materials
Federal-court remand/error rate	60–65%	Agency and oversight data
Incarceration increase after removal	60%	Empirical literature
Mortality elasticity w.r.t. income	−0.6 to −1.0	Gelber et al. (2023)
Projected annual deaths under austerity	∼67,000	Congressional oversight

and a commission's task is precisely to interrogate those assumptions. The book reports the table as the petition offers it, and states the epistemic status of each row.

21.3 The substantive claims

The petition advances claims under four provisions.

Article 6 — the right to life.
The central claim is that the knowing maintenance of an adjudicative system that predictably causes preventable death is itself a breach of the duty to protect life. The Human Rights Committee's General Comment 36 is the interpretive vehicle: it links the right to life to the state's positive obligations and to the conditions necessary for a dignified existence.

Article 9 — liberty and security.
The claim is that claimants subject to multi-year administrative process, restrictive conditions, and repeated monitoring are subject to a continuous deprivation of liberty without a timely determination.

Article 2(3) — an effective remedy.
The claim is that a remedy that remands without deciding is not an effective remedy, because it returns the claimant to the queue that produced the harm.

CAT Article 16 — cruel, inhuman, or degrading treatment.
The claim is that the causal pathway from benefit denial to poverty, homelessness, contact with the criminal-justice system, and incarceration constitutes the prohibited treatment at a structural scale.

21.4 The "restraint of silence" and cognitive accessibility

The neurogenetic dimension is the petition's most distinctive and most contested. The petition argues that the state's categorical refusal to consider validated neurogenetic evidence of an executive-function burden, and its failure to provide a cognitive accommodation, amount to a form of exclusion from the adjudicative process — a "restraint of silence" — that violates the Convention on the Rights of Persons with Disabilities. The strength of the argument lies in its framing: it does not ask the state to accept a contested causal claim about a specific marker; it asks the state to supply an accommodation that would let a cognitively burdened claimant participate. That framing is stronger than the biomarker claim, and the book separates them.

21.5 Requested relief

The petition requests the establishment of a commission of inquiry, the inclusion of the relevant special rapporteurs, a report within twelve months, interim measures suspending workforce reductions, expedited adjudication for high-risk claimants, and a referral to the International Court of Justice for an advisory opinion on the questions stated in Chapter 22.

CHAPTER 22

The Systemic Gravity Index and Treaty Mapping

A metric that measures the gravity of a violation is an argument in numerical form. Its value lies not in its precision but in the questions it forces an institution to answer.

22.1 The index

The Systemic Gravity Index aggregates the petition's quantifications into a single comparative instrument, so that the scale of the alleged harm can be stated in a form an international body can weigh against other matters on its docket.

Table 22.1: Components of the Systemic Gravity Index.

Component	Quantity
Documented mortality	Deaths of claimants awaiting adjudication
Temporal magnitude	Multi-year median processing times
Systemic recurrence	Annual percentage of initial denials reversed on review
Downstream harm	Post-removal incarceration and homelessness
Vulnerability of population	Prevalence of high-mortality conditions in the queue
Institutional knowledge	Evidence of notice and persistence

An index of this kind is a communication device rather than a measurement. Its components are not commensurable, and no weighting can make them so. Its function is to prevent a reviewing body from assessing a systemic harm as if it were an isolated dispute, and in that function it is useful.

22.2 Treaty mapping

The petition maps its claims onto three instruments and one body of customary norms.

ICCPR.
Articles 2(3), 6, and 9 supply the right to a remedy, the right to life, and the right to liberty and security. The interpretive anchor is the Human Rights Committee's General Comment 36.

CAT.
Article 16 supplies the prohibition on cruel, inhuman, or degrading treatment, invoked at the structural rather than the individual scale.

CRPD.
Articles 10, 12, 19, 26, and 28 supply the right to life on an equal basis, the right to legal capacity, the right to live in the community, the right to habilitation and rehabilitation, and the right to an adequate standard of living. The cognitive-accessibility claim in particular is an Article 12 and Article 26 claim.

Jus cogens.
The petition invokes the peremptory character of the prohibition on arbitrary deprivation of life to argue that the obligations are not defeasible by domestic procedure.

22.3 The questions referred

The petition proposes four questions for an advisory opinion. Does the maintenance of an administrative system that predictably causes preventable mortality violate Article 6? Does the deprivation of income, with its documented consequences, constitute treatment within Article 16 of CAT? Is a system whose review mechanism returns a majority of cases to the body that denied them an effective remedy within Article 2(3)? And does the criminalisation of claimants resulting from benefit removal violate Article 9?

The questions are well framed, and that is the chapter's point: a referral is only as useful as the questions it asks, and these questions are answerable in principle even by a body that discounts the petitioner's factual characterisations.

22.4 The limits of the mechanism

Three limits should be stated plainly. International mechanisms are slow, they are non-coercive, and they do not by themselves alter domestic administration. A commission of inquiry can establish facts and name responsibilities; it cannot order a legislature to fund an agency. The petition's value is therefore evidentiary and normative — it creates an authoritative record — rather than remedial. The book states this because the register discipline of Chapter 2 requires it: a request is not a remedy, and a report is not a judgment.

Part VIII

The PSDLEF Sovereign Exchange Trust

CHAPTER 23

Statutory Authority and Juridical Structure

A purpose trust is a legal form that lets an institution outlive both its founders and its beneficiaries. That permanence is the point, and it is also the risk.

23.1 The instrument

The PSDLEF Sovereign Exchange Trust is proposed as a perpetual purpose trust — a trust without individual human beneficiaries, established to carry out a stated purpose in perpetuity. The author frames it under the Uniform Directed Trust Act and the purpose-trust provisions of Delaware and Wyoming law. The stated purpose is the permanent capitalisation, administrative optimisation, and solvency support of federal disability and safety-net programmes.

Register

The purpose trust is an adopted legal form; the trust described here is **[PROPOSAL]**. No such instrument has been created, and the chapter describes the structure the author proposes rather than an executed indenture. Where the chapter reports yields, the figures are **[DESIGN TARGET]**: they are the returns the instrument would need to produce, not returns it has produced.

23.2 Why a perpetual purpose trust

Three features of the form are doing work. First, perpetuity: the trust does not terminate on a beneficiary's death, which suits a purpose measured in decades. Second, non-charitable purpose: the trust is not a charity and is not constrained by charitable-purpose doctrine, which allows a broader investment mandate than a charitable remainder would. Third, directed trusteeship: investment authority is separated from distribution authority, which is the structural protection against a trustee that invests to fund itself rather than to fund the purpose.

The counterweight is that a perpetual purpose trust is precisely the vehicle in which accountability is hardest to maintain, because there is no residuary beneficiary with standing to complain. This is why the governance design of Chapter 24 matters more than the investment design of Chapter 25, and the book orders them accordingly.

23.3 The four-component architecture

The trust is the legal shell for the four technical components the portfolio has already introduced: the MORTAR triage engine (Part V), the OpenSSA integrity pipeline (Part VI), the DYREO capital-allocation optimiser (Chapter 25), and the post-quantum security module (Chapter 20). The claim is that a purpose trust can hold these components as operational assets serving a public purpose, and can fund their operation from the return on a portfolio rather than from appropriations.

23.4 The constitutional question

A trust that performs functions the federal government performs raises two constitutional questions, and the chapter states them rather than resolving them.

1. **The non-delegation and Appointments questions.** Adjudication of disability claims is a governmental function. A private trust may not exercise it. The design therefore confines the trust to capitalisation and to the supply of tools; the agency decides, the trust funds.
2. **The sovereign-immunity question.** A trust that holds assets for a public purpose and is funded by the state may be treated as an arm of the state, which would import sovereign immunity and alter the remedies available. The design mitigates this by keeping the trust private and the state a beneficiary-like recipient rather than an owner.

23.5 The funding rail

The proposed funding rail routes assets through a Wyoming custodial structure into the trust, with multi-signature control and post-quantum signatures on authorisations, under a policy of no leverage, no lending, and no derivatives. The no-leverage policy is the design's most important prudential commitment: a trust whose purpose is solvency cannot itself be a source of systemic risk.

CHAPTER 24

Governance, Fiduciary Roles, and the Perpetuity Rule

The rule that a trust may distribute at most five per cent of its corpus is a rule against the future being consumed by the present.

24.1 Fiduciary separation

The governance design separates four roles. The trust director holds investment authority. A distribution committee holds the authority to release funds to beneficiaries. An independent audit function reviews both. And a standards committee owns the model and policy versions on which the technical systems depend. The separation is not ornamental: the failure mode it prevents is the one in which the entity that benefits from a distribution also decides that the distribution should occur.

24.2 The perpetuity rule

The trust's foundational rule limits distributions to five per cent of a trailing five-year average of the corpus:

$$D(t) \le 0.05 \times \frac{1}{5} \sum_{k=1}^{5} \text{Corpus}(t-k). \tag{24.1}$$

Any investment yield above five per cent is retained and compounded within the capital-preservation core. The rule is the standard spending rule of an endowment, and its function is to make the trust's claims about perpetual returns internally consistent: a trust that distributes more than it earns cannot be perpetual, and the rule makes that impossibility a matter of arithmetic rather than of good faith.

If a distribution rate of five per cent is applied to the trailing corpus and investment return is assumed to be eight per cent nominal, the corpus grows at roughly three per cent nominal, which is a real growth rate of approximately one per cent at two per cent inflation. That is a thin margin, and it is the reason the chapter treats the trust's solvency claim as contingent on the return assumption rather than as established by the structure.

24.3 Drift detection

The design includes a monitoring mechanism that tests whether the distribution of the trust's inputs has shifted away from the baseline on which its models were calibrated. The test is the two-sample Kolmogorov–Smirnov statistic,

$$D_n = \sup_x \left| F_{\text{empirical}}(x) - F_{\text{baseline}}(x) \right|, \tag{24.2}$$

evaluated against a threshold at $p < 0.05$, with a re-optimisation triggered when the threshold is crossed. The mechanism is the governance counterpart of the fairness audit in Chapter 18: it makes a change in the population visible before it becomes a change in outcomes.

24.4 Reporting

The trust reports on a schedule that separates financial from operational performance, and it reports the model versions that produced each operational number. The requirement is drawn from Part VI: a report that does not identify the model that produced it is a report that cannot be audited.

CHAPTER 25

DYREO Capital Allocation and the 31-Algorithm Execution Suite

An optimiser is a statement about what matters. Writing down the objective function is therefore a normative act disguised as a mathematical one.

25.1 The DYREO formulation

The Dynamic Yield and Real Estate Optimization model is a constrained multi-period portfolio problem. In its one-period form it maximises a weighted combination of financial return, social impact, and risk:

$$\max_{w} \; \alpha\, w^{\top}\mu + \beta\, w^{\top}s - \gamma\, w^{\top}\Sigma w, \tag{25.1}$$

subject to a minimum social-impact constraint, a capital-preservation probability constraint, asset-class bounds, and full investment:

$$w^{\top}s \geq S_{\min}, \tag{25.2}$$

$$P(w^{\top}r < -\delta) \leq \epsilon, \tag{25.3}$$

$$w_{\text{housing}} \in [0.40, 0.50], \quad w_{\text{clean}} \in [0.15, 0.25],$$
$$w_{\text{fixed}} \in [0.20, 0.30], \quad w_{\text{liq}} \in [0.05, 0.15], \tag{25.4}$$

$$\sum_{i} w_i = 1, \quad w_i \geq 0 \;\forall i. \tag{25.5}$$

Here w is the weight vector, μ the expected financial yields, s the social- impact scores, Σ the covariance matrix, and α, β, γ the tuning weights. The formulation is a standard mean–variance problem with two additional constraints, and it is honest about its own structure: the social objective enters linearly through s, which means the model treats social impact as commensurable with return, and that commensuration is a choice the model does not justify.

25.2 Asset classes and target yields

The target yields are [DESIGN TARGET]. A portfolio with forty-five per cent of its capital in housing and twenty per cent in clean-energy infrastructure is an illiquid portfolio, and illiquidity is a real cost that the target yields do not show. The chapter's contribution is to state the cost the model omits.

Table 25.1: DYREO asset-class structure and target yields.

Asset class	Allocation	Target yield
Sustainable housing	45%	6.5–8.5%
Clean-energy infrastructure	20%	7.0–9.0%
Sovereign / fixed income	25%	4.5–5.5%
Liquid reserves	10%	4.0–4.5%

25.3 The social energy return

The design measures the social return on the energy assets with a ratio,

$$\text{S-EROI} = \frac{\int_0^T \left[P_{\text{solar}}(t) - P_{\text{loss}}(t)\right] dt}{K_{\text{cap}} + \int_0^T C_{\text{maint}}(t)\, dt} \geq 0.73 \text{ kWh/\$}, \tag{25.6}$$

which compares lifetime energy delivered to lifetime cost. The threshold of 0.73 kWh per dollar is a design target, and its meaning depends entirely on the discount rate and the time horizon, both of which the expression suppresses.

25.4 The 31-algorithm execution suite

The financial infrastructure is specified as a suite of thirty-one algorithms organised into modules.

Table 25.2: Module structure of the executive algorithmic suite.

Module	Function	Indices
I	Bond valuation and claims	01–03
II	Asset conversion and distribution	04–06
III	Volatility modelling	07–09
IV	Execution optimisation	10–12
V	Derivatives and hedging	13–15
VI	Risk management	16–18
VII	Yield optimisation	19–20
VIII	Automated market-maker liquidity	21–22

The module taxonomy is descriptive of the author's library rather than of a proprietary trading system, and the book treats it as such. Its relevance to the argument is that it supplies the operational machinery the trust would need if it were ever funded, and its presence in the portfolio is a reminder that a perpetual purpose trust is an institution with an investment office, not merely a legal form.

CHAPTER 26

Energy and Infrastructure: The Solar/Battery Pilot and Bounded Quantum Research

The responsible way to hold a speculative asset is to size it so that its total loss is survivable.

26.1 The thirty-house solar and battery pilot

The pilot proposes to retrofit thirty Baltimore rowhouses with photovoltaic generation and lithium-iron-phosphate battery storage, forming a decentralised microgrid capable of islanding.

Table 26.1: Scaled thirty-house pilot specification.

Component	Specification
Houses	30 Baltimore rowhomes
Generation	∼5 kW per house, 150 kW total
Storage	∼10 kWh LFP per house, 300 kWh total
Grid model	Decentralised microgrid; 24-hour islanding
Revenue	Energy-service tariffs plus affordable-housing leases
Municipal target	5,000-unit vacancy reduction by FY2029

The pilot is a **[DESIGN TARGET]**. It is small enough to be a genuine demonstration and large enough to test the financing model, and its revenue assumption — that energy-service tariffs and lease income together cover capital and maintenance — is the assumption that must be tested first, because the S-EROI of equation (25.6) is only as good as that assumption.

26.2 Funding rail and control

The proposed rail converts custodial assets into restricted grant funding for the pilot through regulated venues, with multi-signature control and post-quantum signatures. The chapter's editorial position is that the funding rail is the weakest link in the whole financial argument, because it depends on the character of the assets converted and on the regulatory classification of the conversion, and neither is settled by the design.

26.3 The quantum gravitational reactor as a bounded allocation

The specification includes a research allocation labelled the quantum gravitational reactor. The specification itself, to its credit, opens with a critical caveat: no commercially operational device of this kind exists, and the component is a high-risk, optionality-preserving research layer rather than a load-bearing energy asset. The book preserves that framing and strengthens it.

Why the caveat is the correct treatment

The technologies catalogued in the source — pulsed-power radiation sources, theoretical neutron-superfluidity proposals, zero-point-energy transduction, and concept-only gravitational-wave patents — span a range from real research programmes to claims that the mainstream literature regards as unfounded. The correct institutional treatment of a speculative research allocation is not suppression but sizing: capital is committed in an amount whose total loss leaves the mission intact. The design does this, with a success gate of a peer-reviewed paper or an independent replication and a failure mode of a write-down that does not touch the solar and battery assets.

26.4 The due-diligence checklist

The specification's due-diligence questions are worth reproducing because they are the right questions: what is the claimed energy balance; has the effect been independently replicated; is there a peer-reviewed description of the mechanism; what is the theoretical basis; and what are the safety and regulatory requirements. A research allocation that cannot answer all five should not be funded, and the checklist is the trust's defence against its own enthusiasm.

CHAPTER 27

CUSIP Normalization, the Maryland Transfer, and the Dual-Court Filing

Registry mechanics are unglamorous and decisive. An asset that cannot be identified cannot be transferred, and a claim that cannot be filed cannot be heard.

27.1 CUSIP normalization

The protocol normalises securities identifiers under the standard check-digit algorithm, a modified Luhn scheme: convert alphabetic characters to numeric values, double the values at even positions, sum the digits of each product, and take the check digit as $(10 - (P \bmod 10)) \bmod 10$. The algorithm is a standard identifier-validation routine, and its inclusion in the portfolio is a sign that the financial plan is meant to be executed at the level of individual instruments rather than at the level of a diagram.

27.2 The Maryland transfer protocol

The transfer follows the Maryland Uniform Disposition of Abandoned Property Act: identify the identifiers in the portfolio; validate their format and check digit; prepare the transfer documentation, including authorisation, proof of ownership, and taxpayer identification; submit through the state unclaimed-property portal; receive a claim number; and schedule a pickup. The protocol is procedural, and its status is [PROPOSAL] as to the transfer and [CITE REQUIRED] as to any particular asset, because a transfer depends on ownership that the record must establish.

27.3 The dual-court filing

The litigation strategy for the intellectual property and trade aspects is coordinated across two fora.

Court of International Trade.
The trade-administration and transfer aspects are pleaded under the residual jurisdiction of 28 U.S.C. § 1581(i), which reaches civil actions against the United States arising out of federal trade law.

Court of Federal Claims.
The compensation claim for government use of a patented invention is pleaded under 28 U.S.C. § 1498(a), which supplies exclusive jurisdiction over such claims.

Transfer.
Where a court lacks jurisdiction, transfer to cure the defect is requested under 28 U.S.C. § 1631, if in the interest of justice.

27.4 The substantive arguments

Three arguments carry the filing. The first is a public-trust anchoring under the Bayh–Dole Act, which requires that federally supported inventions be utilised to promote free competition and public availability (Bayh–Dole Act 1980). The second is Supremacy Clause preemption of conflicting state assertions, for which the petition cites *United States v. Little Lake Misere Land Co.* (United States v. Little Lake Misere Land Company 1973). The third is the four-factor injunction standard of *Winter v. Natural Resources Defense Council* (Winter v. Natural Resources Defense Council 2008), already used in Part II.

27.5 Requested relief

The filing requests that the court assume jurisdiction, order transfer if warranted, declare the federal rights and the public-trust framework, enjoin state officials from altering cryptographic logs or asserting conflicting claims, direct the recording of federal interests and the implementation of open licensing, and grant leave to proceed in forma pauperis under 28 U.S.C. § 1915. Each request carries its own threshold, and the chapter states them as the filing states them rather than as achievements.

27.6 Forward reference to the integrated capital arithmetic

The CUSIP normalisation described in this chapter is the ledger-side precondition for the capital-structure arithmetic. Chapter 42 (Part XIII) applies the modified Luhn validator to the specific 1999 USEC indenture, runs the 0.0917 plan exchange ratio against the $500M legacy principal, and states the consolidated capitalisation vector on which the DYREO engine operates.

Part IX

Statement of Facts and Procedural History

CHAPTER 28

The Discipline of a Statement of Facts

A statement of facts is not a story. It is a series of propositions, each of which the reader can trace to a document that can be produced.

28.1 Why the record comes last

Parts I–VIII set out a legal theory, a scientific instrument, a computing architecture, and a financing structure. Every one of those parts rests on claims about what has happened to a named person in named forums. This part turns the book's own analytical instruments on the book's own record. It is placed here, after the theory and before the policy, because it can only be written honestly once the reader knows what the theory asks the facts to prove (42 U.S.C. 1983 2018).

The chapter states the events, the forums, and the parties in chronological order. It then marks, in the text itself, every proposition that the record does not yet establish. The marker **[CITE REQUIRED]** is used throughout this part exactly as it was introduced in Chapter 2: it names the document that would close the gap, and it is not a rhetorical hedge. A proposition carrying the marker should not be quoted in a filing until the named document is in the record.

28.2 The three registers of a factual assertion

Facts in litigation are not all of one kind, and the discipline of this part is to keep the kinds apart.

Recorded.
An event appears in a document that a custodian produced: a docket entry, a notice of decision, a transcript, an agency order. A recorded fact is **[ADOPTED LAW]** in the sense that its occurrence is the sort of thing a court will accept from the document itself.

Inferred.
A recorded fact supports a conclusion that goes beyond what the document says. An inference is legitimate only if it is stated as an inference and the competing inference is identified. Most of what a complaint alleges about motive is of this kind.

Asserted.
A proposition appears only in the author's own papers. It is not yet evidence. It is marked **[CITE REQUIRED]** and carries the burden of identifying what would make it evidence.

28.3 The adversarial check

A statement of facts written by a party is tested by asking how the opposing party would answer it. This part therefore states, for each contested event, the adverse fact that a respondent would supply. An inconsistent state-court entry, an adverse administrative finding, or a missed filing deadline is disclosed here rather than omitted. A fact section that omits the adverse entries is not a failing of candour only; it is a failing of strategy, because the omission will be found and will cost the pleader the benefit of the doubt on every other sentence (Heck v. Humphrey 1994).

Every date in this part should come from a docket entry or a notice, never from recollection. Every quotation should carry a pin cite. Every assertion that a government actor did something must identify the actor, the date, the capacity in which the actor acted, and the document that records the act. Where any one of those four elements is missing, the text says so.

CHAPTER 29

The Chronology in Outline

The order of events is itself an argument. A reader who has the chronology does not need to be told which facts matter.

29.1 The forums

The author's matters have proceeded, or are described as proceeding, in four different forums. Keeping them distinct is a precondition of coherent analysis, because the doctrines, the standards of review, and the available remedies do not transfer between them.

Table 29.1: The forums described in the project materials.

Forum	Route	What it can decide
Federal district court	42 U.S.C. § 1983	Civil-rights claims against state and local officials (42 U.S.C. 1983 2018)
Social Security review	42 U.S.C. § 405(g)	Judicial review of a final agency decision on disability benefits (42 U.S.C. 405(g) 2018)
Court of Federal Claims	28 U.S.C. § 1491	Money claims against the United States, including patent claims under § 1498 (28 U.S.C. 1491 2018)
State court / bankruptcy court	Case-specific	The underlying state matter and the Chapter 11 proceeding (Bankruptcy Code 1978)

29.2 The federal civil-rights matter

The project materials describe a federal civil-rights action assigned the docket numbers 25-1222, 25-1224, 25-1522, and 24-3283, arising from interactions with state and local officials in Maryland. The materials characterise the matter as presenting claims for retaliatory arrest, coercive interrogation, fabricated evidence, conspiracy, and municipal liability — the five counts analysed doctrinally in Part II. The factual predicate for each count is a discrete encounter; the source materials identify and date several of these encounters. **[CITE REQUIRED]** The operative complaint, the docket sheets, and the transcripts of the relevant proceedings would convert the description into a record.

Two limitations must be stated plainly. First, the existence of a state criminal matter does not by itself establish any federal constitutional violation. Each inference from the state matter to a federal claim requires its own factual and legal basis. Second, the standard for a retaliatory-arrest claim after *Nieves v. Bartlett* and its 2024 refinement in *Gonzalez v. Trevino* is demanding: the plaintiff must plead and prove both a

retaliatory motive and the absence of probable cause, or else fall within a narrow class of cases otherwise (Nieves v. Bartlett 2019; Gonzalez v. Trevino 2024). A pleading that recites retaliation without confronting the probable-cause element invites dismissal.

29.3 The Social Security Administration matter

The materials describe a disability matter before the Social Security Administration and a draft petition, dated March 15, 2026, invoking 42 U.S.C. § 405(g) against the Administration, the Department of Health and Human Services, and the United States. **[CITE REQUIRED]** The file-stamped petition, the final agency decision, and the administrative-record index are needed to fix the parties, the decision under review, the statutory route, and the relief sought.

The complete history of such a matter has a settled shape, and the record must supply each stage or the stage must be omitted: initial application; initial determination; reconsideration, if any; hearing request; the administrative law judge's decision; Appeals Council action; the dates of each notice; and any subsequent judicial filing. Every one of those dates should come from a notice or a docket entry rather than from inference.

Register

That the Social Security disability queue is long, and that claimants have died while appeals were pending, is **[ADOPTED LAW]** on the strength of official reporting (U.S. Government Accountability Office 2020). That a specific claimant's matter followed a specific procedural path is a matter of **[CITE REQUIRED]** until the notices are in the record. That a given triage design would have shortened that claimant's wait by a stated number of days is a **[MODEL / SIMULATION]** proposition.

29.4 The bankruptcy and securities background

The Chapter 11 materials describe a personal reorganisation whose principal asset is a portfolio of securities reportable under Section 13(f) of the Securities Exchange Act (Securities Exchange Act of 1934 1934). The plan classifies claims into the ordinary administrative, priority, secured, and unsecured classes and proposes distribution from the liquidation of the portfolio (Bankruptcy Code 1978). Because the estate is an individual's and the portfolio is the funding source for everything else the book proposes, the liquidation schedule is treated in Part VIII as the hinge of the financing structure. **[CITE REQUIRED]** The schedules, the statements of financial affairs, and the liquidation analysis would establish the values the plan relies upon.

A Subchapter V reorganisation, if available on the facts, alters the procedural posture — it dispenses with the creditors' committee, fixes a short exclusive period for the debtor to propose a plan, and permits the debtor to retain equity over an objection in defined circumstances (Small Business Reorganization Act of 2019 2019). Whether the debtor qualifies is a question of statutory thresholds that the record must answer.

CHAPTER 30

Technical and Scientific Evidence in the Record

An assertion becomes evidence when someone can say what was measured, with what instrument, by whose hand, and with what result.

30.1 Genomic and neurophysiological evidence

The materials propose a framework that fuses functional evidence with genomic, neuroimaging, and neurobiological information. They refer to candidate markers, among them COMT rs4680 and loci associated with executive function, together with crosswalks to Social Security Listings 11.00, 12.00, and 12.11. Those references do not establish that the Administration received the underlying genotyping, that a qualified examiner interpreted it, or that any adjudicator relied on it. **[CITE REQUIRED]** The laboratory reports, the chain of custody for the samples, and the treating-source records would establish what was measured.

The evidentiary status of a polygenic score is governed by Part III. A score is a population-level statistical estimate; its probative value in an individual case depends on the score's construction, its validation population, and its calibration. Under *Daubert* and Rule 702, the proponent must show that the method is reliable and that it has been reliably applied, and a court may exclude a score that its own author concedes is not diagnostic (Daubert v. Merrell Dow Pharmaceuticals 1993; Federal Rule of Evidence 702 2023; Kumho Tire Company v. Carmichael 1999).

30.2 The ledger, the pipeline, and their limits

The materials describe a hash-chained ledger and a provenance pipeline — the OpenSSA Integrity Pipeline — intended to make a sequence of records tamper-evident. A hash chain can establish that a recorded sequence has not been altered since a root was published, under stated assumptions (Merkle 1979; Haber and Stornetta 1991). It does not, by itself, prove that the underlying assertion is true, that the record was complete, or that the entry was admissible.

Source authenticity. A chain proves internal consistency, not that the first entry described the world accurately.
Truth of the assertion. A signed record of an untrue statement is still an untrue statement.
Legal admissibility. Self-authentication addresses the form of the record, not its relevance or its

probative weight.

If a ledger is offered as proof of a historical event, the proponent must identify exactly what it records and why the record is reliable: how entries entered the system, who controlled the keys, how timestamps were generated, how corrections were recorded, and how the implementation was tested. **[CITE REQUIRED]** The repository snapshot, the dependency inventory, the test results, and any independent assessment would establish the technical claims.

30.3 Cryptographic claims

The materials refer to FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA), and to NSA's CNSA 2.0 suite (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d; National Security Agency 2022b). Mentioning a standard does not prove conformance with it. Any claim that a component implements one of these algorithms should identify the actual library, the parameter set, the test vectors, the key-management process, and any independent validation such as a FIPS 140-3 certificate (National Institute of Standards and Technology 2019). Where a component uses a placeholder, an HMAC, or a simulated primitive, the text of this part says so, and the corresponding claims in Parts IV and VI are read as design targets rather than as measurements.

30.4 HARMONI-X and the cognitive-accessibility record

HARMONI-X is described as a proposed assistant for document organisation, omission alerts, and cognitive support. A tool can flag a missing keyword or a cross-reference; the alert does not establish that a legal omission occurred. The record of a cognitive-accessibility claim should state what was checked, what a human reviewer concluded, and what the material limitations were. Where an accommodation is claimed to have been sought, the request, its date, its recipient, and the response — or the absence of a response — should appear in the record. If no accommodation request appears on the docket, the fact statement must not say that the court denied one.

CHAPTER 31

Gaps, Adverse Facts, and the Verification Protocol

The most useful page in a fact section is the one that lists what the pleader cannot yet prove.

31.1 The catalogue of gaps

The following propositions are asserted in the project materials but are not, on the present record, established. Each is stated here with the document that would close it. The list is not exhaustive, and it is meant to shrink as the record is assembled.

1. That a specific federal complaint was filed and served on named defendants in 25-1222, 25-1224, 25-1522, or 24-3283. **[CITE REQUIRED]** Operative complaints and docket sheets.

2. That a retaliatory motive existed and that probable cause was absent in any particular arrest. **[CITE REQUIRED]** Charging documents, arrest reports, and the disposition of any criminal charge (Nieves v. Bartlett 2019).

3. That the Social Security matter reached a final decision reviewable under § 405(g), and the date of that decision. **[CITE REQUIRED]** The notice of decision and the Appeals Council order (42 U.S.C. 405(g) 2018).

4. That a particular polygenic score was validated at the level asserted, and that it was before the adjudicator. **[CITE REQUIRED]** Validation study, laboratory report, and the administrative exhibit list (Daubert v. Merrell Dow Pharmaceuticals 1993).

5. That any component of the pipeline uses a real post-quantum implementation rather than a placeholder. **[CITE REQUIRED]** Repository dependency manifest and test transcripts (National Institute of Standards and Technology 2024b).

31.2 The adverse facts a respondent would supply

A fact section must anticipate the other side. Three adverse facts bear on the narration above and are disclosed here.

The first is that an unresolved state-court matter may bar or delay the federal civil-rights claims under the favourable-termination requirement, which in most circuits is satisfied only by a termination that affirmatively indicates innocence, and which prevents a § 1983 damages claim that would impugn an

outstanding conviction (Heck v. Humphrey 1994; Thompson v. Clark 2022). A stay under the abstention doctrine may follow if the state proceeding is ongoing and involves important state interests (Younger v. Harris 1971).

The second is that several of the asserted technical facts are, on the face of the materials, aspirational: the architecture is a specification, the quantum substrate is simulated, and the cryptographic module is described at the level of interfaces. A respondent would say that no working system has been shown to exist. The register discipline of this book is designed to answer that objection before it is made, by labelling each claim.

The third is that the damage figures used to motivate the triage engine are order-of-magnitude estimates drawn from official reporting, not measurements of the author’s own cohort (U.S. Government Accountability Office 2020; Sanders and Larson 2020). They are reported as estimates and must not be presented as findings about any individual.

Part X

The Integrated Policy Framework

CHAPTER 32

From Case to Policy: The Gov-LLM Proposal

A remedy that helps one claimant is a judgment. A remedy that helps the next claimant is an institution. This part is about the second.

32.1 The proposition

The legal analysis of Part II and the evidentiary analysis of Part III converge on a single institutional diagnosis: the agencies that decide whether a disabled person receives support are asked to apply a body of law too complex for any individual adjudicator to hold in memory, to a record too heterogeneous to be compared across cases, on a timetable too short to be fair. The proposal that follows from that diagnosis is a state-government legal-knowledge system — referred to in the project materials as a Gov-LLM — embedded in a benefits and accommodation workflow.

The idea is not to let a model decide eligibility. It is to let a model perform three bounded functions that adjudicators now perform badly under time pressure: retrieving the governing authority for a given factual pattern, flagging a record that is missing a document the authority requires, and drafting the boilerplate that surrounds a human decision. Every one of those functions is **[PROPOSAL]**. The decision itself remains a human act.

Register of this part

Nothing in this part is adopted law. The federal framework described below is **[ADOPTED LAW]** where it names a statute, a rule, a published standard, or a reported decision; it is **[PROPOSAL]** where it recommends an institutional reform; and it is **[MODEL / SIMULATION]** where it reports a performance figure from a prototype. A reader who wants to know whether a paragraph can be relied upon in a brief should read the register first.

32.2 Why a knowledge system and not a decision system

The distinction is constitutional, not merely prudential. Adjudication of disability benefits is an exercise of the state's coercive authority over property and liberty interests, and it attracts procedural due process (Mathews v. Eldridge 1976). *Mathews v. Eldridge* fixes a three-factor balance: the private interest at stake, the risk that the existing procedure will produce an erroneous deprivation, and the government's interest in the added burden of a more protective procedure. A knowledge system can be justified under the second factor — it reduces the risk of error — without ever being asked to supply the decision that the first and third factors protect.

32.3 The deployment shape

The proposed system is described as running on standard agency hardware, including laptops, with a short deployment time. That is a claim about resource footprint rather than about accuracy, and it is **[MODEL / SIMULATION]** where it rests on a prototype and **[DESIGN TARGET]** where it rests on design intent. The relevant legal question is not the footprint but the auditability of the output: an agency that cannot reproduce the retrieval that produced a recommendation cannot defend the recommendation on review.

CHAPTER 33

The Statutory Lattice of a State Legal-Knowledge System

The law an agency must apply is not one statute. It is a lattice of overlapping duties, each with its own remedy and its own forum.

33.1 The disability and education core

A system that supports accommodation and benefits decisions must be indexed to a body of law that spans at least three fields. In disability and education, the governing instruments are the Americans with Disabilities Act and its 2008 amendments, which define disability in functional rather than diagnostic terms and reject the consideration of mitigating measures in the threshold inquiry (Americans with Disabilities Act of 1990 1990; ADA Amendments Act of 2008 2008); Section 504 of the Rehabilitation Act, which supplies the nondiscrimination duty for programmes receiving federal funds (29 U.S.C. 794 2018); and the Individuals with Disabilities Education Act, which governs the special-education entitlement (Individuals with Disabilities Education Act 2004). FERPA governs the education records that a school-based system would ingest (Family Educational Rights and Privacy Act 2018).

33.2 The benefits core

In the benefits field, the Social Security Act and its disability provisions supply the substantive standard, and § 405(g) supplies the exclusive route for judicial review of a final decision (Social Security Act 1935; 42 U.S.C. 405(g) 2018). The procedural-due-process framework of *Mathews* and its application to benefits terminations govern the process the agency owes before it acts (Mathews v. Eldridge 1976).

33.3 The privacy core

Any system that ingests a claimant's health, education, and genetic information touches the privacy lattice as well. HIPAA and the Security Rule govern protected health information held by covered entities and their business associates (Health Insurance Portability and Accountability Act of 1996 1996). GINA prohibits the use of genetic information in health insurance and employment but, as Part II emphasises, leaves life, disability, and long-term-care insurance largely unaddressed (Genetic Information Nondiscrimination Act of 2008 2008). The gap is the reason the genomic-justice pillar of the directive described in Chapter 35 is posed as a legislative proposal rather than as an enforcement action.

Table 33.1: The statutory lattice and the agency that interprets it.

Instrument	Key duty	Interpreting authority
ADA / ADAAA	Nondiscrimination; reasonable accommodation	EEOC, DOJ, courts (Americans with Disabilities Act of 1990 1990)
Rehabilitation Act § 504	Nondiscrimination in federally funded programmes	Federal agencies, OCR (29 U.S.C. 794 2018)
IDEA	Free appropriate public education	OSEP (Individuals with Disabilities Education Act 2004)
FERPA	Confidentiality of education records	Department of Education (Family Educational Rights and Privacy Act 2018)
Social Security Act	Disability benefits	SSA (Social Security Act 1935)
HIPAA / Security Rule	Privacy and security of health information	HHS, OCR (Health Insurance Portability and Accountability Act of 1996 1996)
GINA	Genetic-information nondiscrimination (health insurance, employment)	EEOC, HHS (Genetic Information Nondiscrimination Act of 2008 2008)
APA	Procedural regularity in agency action	Courts (Administrative Procedure Act 1946)

33.4 The drafting consequence

The lattice has a practical consequence for the design of any legal-knowledge system: authority cannot be stored as a flat list of documents. It must be stored as a graph in which a factual pattern resolves to a duty, the duty resolves to the agency that interprets it, and the interpretation resolves to the remedy and the forum. That requirement is precisely the architectural thesis of Part IV, which is why the semantic-execution graph and the statutory lattice are two descriptions of one structure.

CHAPTER 34

Algorithmic Governance and the Federal–State Divergence

In three years the federal stance toward AI regulation moved from mandated risk management to mandated deregulation, while the states moved the other way. An agency designing a system must survive both.

34.1 The federal trajectory

The federal posture toward artificial intelligence has reversed twice in three years, and the reversal is itself a design constraint. The National Institute of Standards and Technology published the AI Risk Management Framework in January 2023, organised around the functions *govern*, *map*, *measure*, and *manage* (National Institute of Standards and Technology 2023). It was followed in July 2024 by a generative-AI profile that catalogues risks specific to generative models (National Institute of Standards and Technology 2024a). In the same period the Office of Management and Budget issued Memorandum M-24-10, which required federal agencies to designate a chief AI officer, to inventory their AI use cases, and to apply minimum risk-management practices to rights-impacting and safety-impacting systems (Office of Management and Budget 2024).

Executive Order 14110, issued in October 2023, had made that framework binding across the executive branch and had directed attention to civil-rights implications (The White House 2023). It was revoked in January 2025 by Executive Order 14179, which framed prior requirements as barriers to leadership (The White House 2025a), and the administration's subsequent action plan recast the federal role as accelerating deployment and reducing regulatory burden (The White House 2025b).

A risk-management framework published as a voluntary standard, and a procurement memorandum that agencies have already implemented, survive a change of administration in a way that an executive order does not. For an institution that must remain compliant across political cycles, the durable anchors are the NIST framework and the agency's own procurement rules, not the order of the day.

34.2 The state trajectory

The states have moved in the opposite direction. California's Transparency in Frontier Artificial Intelligence Act, signed in September 2025, imposes transparency and incident-reporting duties on developers of the largest frontier models and protects whistleblowers (California State Legislature 2025). Companion statutes establish independent verification organisations and an auditor registry (California State Legislature 2026b; California State Legislature 2026a). Colorado's 2024 Act imposed a duty of reasonable care against algorithmic discrimination in consequential decisions, though it was later replaced by a disclosure-centred framework with a delayed effective date (Colorado General Assembly 2024).

The divergence produces a genuine compliance problem. A state agency that operates a Gov-LLM is subject to state AI law even where federal law has been relaxed, and it is subject to federal antidiscrimination law and the *Mathews* balance regardless of either. The safe design is therefore the stricter of the two regimes, documented to the NIST framework's own vocabulary so that the documentation can be reused in whichever forum asks for it.

34.3 The civil-rights overlay

Algorithmic governance in the disability context is not only an administrative question. Where a state uses an automated tool to triage or screen benefit applicants, the tool's disparate effect on a protected class is actionable under the same statutes that govern manual decisions, and the integration mandate of *Olmstead* constrains a state's ability to deliver services in institutional rather than community settings (Olmstead v. L.C. ex rel. Zimring 1999). A design that reduces a queue by concentrating error on the most impaired claimants fails both tests at once.

CHAPTER 35

The Three-Pillar Directive and Its Implementation

A directive that names three pillars but no enforcement mechanism is a wish. The three pillars are set out here with the mechanism attached to each.

35.1 The instrument

The materials propose an international policy directive, designated *UN/PSDLEF/GFQA/2026/001*, organised around three pillars that correspond to the three technical parts of this book. The directive is **[PROPOSAL]**. It is useful, however, precisely because it forces the technical, legal, and financial proposals into a single architecture and exposes where they depend on one another.

Table 35.1: The three pillars and the mechanism attached to each.

Pillar	Instruments	Enforcement mechanism
Cognitive accessibility	Cognitive-load modulation, predictable interfaces, assistive software, cognitive-liberty safeguards	ADA Title II and § 504 programme duties (Americans with Disabilities Act of 1990 1990; 29 U.S.C. 794 2018)
Genomic justice	Genomic sovereignty, dynamic consent, data minimisation, algorithmic-discrimination controls, evidence provenance	GINA and HIPAA for the covered sectors; new legislation for the uncovered ones (Genetic Information Nondiscrimination Act of 2008 2008; Health Insurance Portability and Accountability Act of 1996 1996)
Quantum-secure infrastructure	ML-KEM, ML-DSA, SLH-DSA, zero-trust architecture, secure biological-data nodes	Procurement mandates keyed to the CNSA 2.0 transition timetable (National Security Agency 2022b; National Institute of Standards and Technology 2020)

35.2 Cognitive accessibility as the first pillar

The first pillar converts the accessibility requirements of Part II into interface requirements. Where a person's impairment is executive rather than sensory, accessibility is a matter of cognitive load: how many

steps a task requires, how much working memory it demands, and whether the interface behaves predictably across sessions. The legal hook is the accommodation duty itself, which obliges a public entity to make reasonable modifications to its programmes (Americans with Disabilities Act of 1990 1990). The technical hook is the cognitive-interface layer of the architecture in Part IV.

35.3 Genomic justice as the second pillar

The second pillar addresses the asymmetry identified in Part II: the genome is the most identifying and least changeable record a person has, and the statutory protection is fragmentary. The pillar proposes dynamic consent, data minimisation, and provenance controls, and it is candid that the insurance gap cannot be closed by interpretation. It requires legislation. Until that legislation exists, the pillar's operational content is procedural: it requires that any use of genomic information in an adjudicative setting be documented, so that the use is at least visible (Genetic Information Nondiscrimination Act of 2008 2008).

35.4 Quantum security as the third pillar

The third pillar is the most concrete, because it rests on published standards with a published migration timetable. The three post-quantum standards — ML-KEM for key encapsulation, ML-DSA and SLH-DSA for signatures — were published in August 2024 (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d), and the national-security community has set a transition schedule for migrating to them (National Security Agency 2022b). A state system that holds genomic records with a decades-long retention period should be migrating now, because a record encrypted today must remain confidential for longer than the shelf life of the classical cryptography that protects it. Zero-trust architecture supplies the complementary access model (National Institute of Standards and Technology 2020).

35.5 The limits of a policy proposal

Three limits should be stated. The first is jurisdictional: a directive adopted by an international body does not bind a state agency, and a state agency does not implement a treaty. The directive's operative content in any given forum is whatever domestic law already says. The second is fiscal: every pillar has a cost and the book does not pretend that the trust structure of Part VIII can fund all of it. The third is epistemic: the performance figures that motivate the proposal are order-of-magnitude estimates and prototype measurements, labelled as such, and they should not be cited as if they were results of a controlled trial. A proposal that overstates its evidence loses the argument it was made to win.

Part XI

Applied Projects and Capstones

CHAPTER 36

The UMGC Integrated Capstone: Design and Validation Plan

A capstone is judged not by the grandeur of its subject but by whether the student can say what would falsify the design.

36.1 The programme and the brief

The University of Maryland Global Campus integrated capstone joins a degree in biotechnology with one in cybersecurity technology. The manuscript submitted in June 2026 is titled *Integrated Cyber-Biotechnology: Closed-Loop Neuroprosthetics with Post-Quantum Genomic Adjudication*. Its subject is a closed-loop neuroprosthetic — abbreviated CINP in the manuscript — that unites genome-editing machinery with post-quantum cryptographic protection to address executive dysfunction of neurogenetic origin.

The manuscript is a professional degree submission, and this chapter treats it as one: a design with a validation plan, not a report of a completed clinical intervention. Every engineering claim below is **[DESIGN TARGET]** unless the manuscript itself reports a measurement.

36.2 The system as designed

The system has three coupled subsystems, and the coupling is the intellectual point of the capstone.

The sensing layer
derives a real-time polygenic risk score from a genotyping panel. The score is a population-level estimate; the manuscript is explicit that its use in an individual closed loop would require prospective validation, which is the same caution that Part III applies to polygenic evidence in adjudication.

The actuation layer
proposes CRISPR activation and interference modules for on-demand epigenomic control. The manuscript presents these as a design target and identifies the off-target, delivery, and reversibility problems that any clinical translation would have to solve.

The custody layer
protects every genomic, clinical, and legal record with a quantum-resistant audit ledger, so that the data governing the loop cannot be altered without detection. This layer is the part of the design that is engineering-complete, because it depends only on published standards.

Register

That the standards underlying the custody layer are published is **[ADOPTED LAW]** (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b). That the sensing and actuation layers can be built as described is **[DESIGN TARGET]**, resting on a literature review and a design argument rather than on a prototype. That the integrated loop would improve executive function in any individual is neither asserted nor established, and the manuscript does not assert it.

36.3 The validation plan

The manuscript's contribution to the degree, and to the reader of this book, is its validation plan: the sequence of experiments that would falsify each layer before any human subject is exposed. The plan separates analytical validation (does the assay measure what it claims), clinical validation (does the measurement predict the outcome), and utility (does acting on the prediction change the outcome). A capstone that states those three steps in order is a responsible document even when its subject matter is speculative.

CHAPTER 37

The Baltimore Secure Backbone System

Genomic data has a retention period measured in generations. The cryptography that protects it must be chosen for that horizon, not for this year.

37.1 Why a network belongs in this book

The project materials include a municipal network design, the Baltimore Secure Backbone System (BSBS), whose stated purpose is to carry genomic and clinical records under a zero-trust model with mandated post-quantum cryptography. It belongs in this book because it converts the abstract security requirements of Parts IV and VI into a concrete deployment: a place where the cryptographic claims become procurement decisions.

The design is one instantiation of the model published as NIST Special Publication 800-207, in which no actor is trusted by virtue of network location and every request is authenticated and authorised explicitly (National Institute of Standards and Technology 2020).

37.2 The three tiers

The design is organised into three tiers, each with its own trust assumptions.

37.3 The audit interface

The design specifies a small API whose function is to make the audit trail reproducible rather than merely present. In outline, the interface exposes a liveness and handshake-status endpoint, an append endpoint that signs an event into the ledger, a retrieval endpoint that returns an event together with its Merkle proof, a root endpoint that publishes the current Merkle root under an ML-DSA signature, a query endpoint that re-derives a hash-chain segment, and a metrics endpoint in a standard operations format.

The fourth of these — the ability to re-derive a segment on demand — is the one that matters for litigation, because it is what makes the evidentiary claim of Chapter 19 testable. A ledger whose contents cannot be re-derived and compared is an assertion; a ledger with a reproducible derivation is evidence.

37.4 The honest limits of the design

The design is a plan. Three limits follow. First, key custody is the whole of the matter: a zero-trust network whose signing keys are held in one administrator's account is not zero-trust. Second, the migration to

Table 37.1: The three-tier architecture of the BSBS design.

Tier	Scale	Protection
Edge	∼120 sites (hospitals, clinics, homes, laboratories)	SD-WAN tunnels keyed with ML-KEM (National Institute of Standards and Technology 2024c)
Core	Two redundant municipal data centres	Service mesh with workload identity; all east–west traffic ML-KEM-protected (National Institute of Standards and Technology 2020)
Custodial	Hardware-security-module enclaves for genomic and clinical archives	ML-DSA signing keys in FIPS 140-3 Level 3 modules with quorum release (National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2019)

post-quantum primitives requires replacing every handshake and every certificate, and a partial migration protects nothing that traverses an unprotected hop. Third, a hash-chained ledger of records does not make the records true, as Chapter 30 states; it makes them unaltered.

CHAPTER 38

The United Nations Submission

Treaty bodies do not award damages. They establish that a condition exists and name it. Naming is the first remedy.

38.1 The instrument and its venue

In June 2026 the project materials describe a submission to the United Nations Human Rights Council, under the symbol *A/HRC/58/NGO/2026/Rev.2*, presented through the Office of the High Commissioner for Human Rights. The submission is an advocacy document, not a pleading, and this chapter reads it as such.

38.2 The factual claim

The submission's factual core is a mortality claim: that a substantial number of claimants die each year while their disability claims are pending. That the disability queue is long, and that claimants have died while awaiting adjudication, is **[ADOPTED LAW]** on the strength of official reporting (U.S. Government Accountability Office 2020). The submission's specific annual figure is an order-of-magnitude estimate consistent with the reported range, and it is reported in Part V as such (Sanders and Larson 2020). The submission's description of the backlog's scale and of the increase in average wait times over the preceding five years is likewise drawn from official sources and should be cited to them rather than to the submission itself.

Register

That official reporting documents deaths during the wait is **[ADOPTED LAW]** (U.S. Government Accountability Office 2020). That the author's larger figure is accurate to the person is **[MODEL / SIMULATION]** and is not claimed. That an international body should open a commission of inquiry is **[PROPOSAL]**; whether it will is not a matter this book can predict.

38.3 The legal frame

The submission invokes three treaty instruments. The first is the International Covenant on Civil and Political Rights, whose protection of life is non-derogable and whose Human Rights Committee has stated that the obligation extends to deprivations caused by state institutions rather than by deliberate acts alone (International Covenant on Civil and Political Rights 1966; UN Human Rights Committee 2018). The second is the Convention against Torture, whose relevance the submission argues through the concept of administrative suffering that rises to the threshold of cruel, inhuman, or degrading treatment (Convention

against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment 1984). The third is the Convention on the Rights of Persons with Disabilities, whose provisions on accessibility, equal recognition before the law, independent living, and habilitation and rehabilitation supply the affirmative framework (Convention on the Rights of Persons with Disabilities 2006).

The relational argument that connects the three instruments is developed in Part VII and is not repeated here. What this chapter adds is the procedural posture: a submission under the Council's institution-building resolution (UN Human Rights Council 2007) creates a record and a set of named questions, and its value lies in the permanence of the record rather than in immediate relief.

38.4 What the submission does not claim

The submission does not claim that the United States has consented to an individual complaint under an optional protocol that would create a binding outcome, and it does not claim that a treaty body can order relief. It claims that the condition exists, that it is documented, and that it should be investigated. Stated that way, the submission is defensible on its face; stated more grandly, it would not be.

CHAPTER 39

The Chapter 11 Plan of Reorganisation

A financing structure is only as good as the honesty of its schedules. The plan is the instrument, and the schedules are the truth of it.

39.1 The plan and its purpose

The project materials include a Chapter 11 plan of reorganisation filed by the debtor as an individual, whose stated purpose is to resolve outstanding obligations and to provide for an orderly distribution to creditors from the liquidation of a securities portfolio reportable under Section 13(f) of the Securities Exchange Act (Securities Exchange Act of 1934 1934; Bankruptcy Code 1978). Within this book the plan is not a separate matter; it is the funding rail of Part VIII. The trust that the book proposes can only receive assets that the estate can lawfully transfer, and the plan determines what those assets are.

39.2 The classification of claims

The plan follows the ordinary architecture of Chapter 11. It classifies claims and interests into administrative-expense claims and priority tax claims (Bankruptcy Code 1978); secured claims to the extent of the value of the collateral; general unsecured claims; and the debtor's residual interest. It provides for the treatment of executory contracts and unexpired leases, and it fixes the effective date as the first business day after the confirmation order becomes final and non-appealable.

39.3 Discharge and the retention of jurisdiction

On the effective date the plan discharges debts arising before the order for relief to the extent the Bankruptcy Code provides, and the court retains jurisdiction to interpret and enforce the plan. Both provisions are standard, and their significance here is that they set the boundary of the estate: what is discharged is not available to the trust, and what the court retains is subject to its continuing supervision.

39.4 The Subchapter V question

If the debtor qualifies, Subchapter V of Chapter 11 offers a materially different procedure: no creditors' committee, a short exclusive period for the debtor to propose a plan, and the possibility of retaining equity over an objection in defined circumstances (Small Business Reorganization Act of 2019 2019). The qualification thresholds are statutory and must be tested against the schedules, not assumed. Whether to

proceed under Subchapter V is therefore a question of fact about the estate, and the plan's own schedules are the only documents that can answer it.

39.5 The schedules as the hinge of the whole book

Every financial proposition in Part VIII — the perpetuity rule, the payout ratio, the allocation bands — is computed from the corpus, and the corpus is defined by the schedules. If the schedules overstate the portfolio, every downstream figure is wrong in the same direction. The discipline that this book applies to factual assertions in Part IX applies with equal force here: the schedules are **[CITE REQUIRED]** until they are filed and tested, and the financing structure is **[PROPOSAL]** until they are.

Part XII

Conclusion

CHAPTER 40

The Constitutional Infrastructure

The claim of this book is not that the systems it describes are built. It is that the institutions which decide whether a person eats are governed by nothing that was designed for the data they now hold.

40.1 The single argument, restated

The book set out to join four literatures, and it closes by stating what joins them. The joining claim is this: the institutions that hold genomic, neurophysiological, educational, and financial records about a person are constrained, if at all, by instruments drafted for a world in which such records were either impossible or expensive. The genome is the clearest case. It is unchangeable, it identifies without being intended to identify, it implicates relatives who never consented, and the statute that governs it (Genetic Information Nondiscrimination Act of 2008 2008) reaches health insurance and employment and stops there. The consequence is not a single wrong. It is a structural under-protection that appears as a civil-rights claim in one forum, as an evidentiary problem in another, and as a queue in a third.

40.2 The four registers as the book's method

The book's methodological contribution is the register discipline of Chapter 2 and its application throughout. Every proposition in this volume sits on one of four registers: adopted law, proposal, model or simulation, and design target. A claim that the disability queue kills people is adopted law, resting on official reporting (U.S. Government Accountability Office 2020). A claim that a triage engine would shorten the queue to a stated figure is a model. A claim that an agency should adopt such an engine is a proposal. A claim that the quantum substrate of Part IV can be built as specified is a design target. A reader who can place a sentence on its register can decide what to do with it, and that is the test the book sets for itself.

40.3 Why the architecture and the jurisprudence are one thing

A reader may ask why a book about constitutional litigation contains a processor specification. The answer is that the legal argument requires exactly the property the architecture supplies. A constitutional claim that a genomic record has been misused depends on being able to show what was recorded, when, by whom, and with what authority — and on being able to show that the record has not been altered since. That is the definition of provenance, and provenance is what a hash-chained ledger provides. The semantic-execution graph of Part IV is the computational form of the same idea: a record whose meaning travels with it and

whose derivation can be replayed. The litigation theory and the architecture are not two projects; they are the legal and technical descriptions of one requirement.

40.4 The financing claim and its limit

The book's fourth literature is finance, and its claim there is that public-benefit infrastructure can be capitalised through a perpetual purpose trust whose corpus is professionally managed and whose payout is capped by a rule rather than by a board's discretion (Bankruptcy Code 1978). The claim is not that such a structure is novel; it is that the structures that exist are under-used for this purpose. The limit is that a trust cannot receive what the estate does not own, and so the entire financing argument depends on the schedules discussed in Chapter 39. A structure built on overstated assets is not a structure; it is a liability.

CHAPTER 41

What Would Have to Be True

A conclusion that does not state what would falsify it is not a conclusion. It is a preference.

41.1 The falsifiers

The book's claims become knowledge only if they are capable of being wrong. The following are the conditions under which each would fail.

The constitutional claim.
It fails if the courts hold that the collection and use of genomic information by a state agency raises no Fourth Amendment interest beyond the interests already recognised (Carpenter v. United States 2018; Maryland v. King 2013), and that GINA's coverage is the full extent of the legislative protection the legislature intends to give. The first half of that condition is contested and the second is, on the current statute, true.

The evidentiary claim.
It fails if polygenic scores and neuroimaging endophenotypes cannot be shown to be reliably applied to an individual (Daubert v. Merrell Dow Pharmaceuticals 1993; Kumho Tire Company v. Carmichael 1999). That is a live possibility, and Part III treats it as the controlling caution rather than as a settled premise.

The architectural claim.
It fails if the semantic-execution model cannot be shown to improve on a conventional pipeline for the workloads it targets, or if the quantum substrate cannot be built at the scale the specification assumes. The burden here is an engineering burden, and it has not yet been discharged.

The governance claim.
It fails if automated triage systems are shown to reduce error without concentrating it on protected classes (Olmstead v. L.C. ex rel. Zimring 1999). If that were shown, the argument for the triage engine would be stronger, not weaker; the point is that the claim is testable.

The financing claim.
It fails if the schedules do not support the corpus, or if the perpetuity rule cannot be reconciled with the applicable rule against perpetuities in the relevant jurisdiction. The second is a drafting question with a known answer; the first is a question of fact about the estate.

41.2 The next document

The book ends where it began, with a document. The theory of Part II becomes a pleading when the operative complaint is filed. The evidence of Part III becomes a record when the laboratory report is exhibited. The architecture of Part IV becomes a system when the first test vector is published with a hash. The trust of Part VIII becomes a corpus when the schedules are filed.

The institutions that decide whether a disabled person receives support should be required to record what they decide, in a form that can be replayed, under a standard that a court can review — and the technical means to do so now exist, while the legal duty to do so does not.

That is the constitutional infrastructure the title promises. It is not a building; it is a duty. Whether it is imposed is a question for legislatures and courts, and this book has stated, as carefully as it could, what the record already shows and what it does not.

Part XIII

ADR-DYREO-PSDLEF: Ph.D.-Level Quantitative Synthesis

CHAPTER 42

Jurisdiction, Due Process, and Cognitive Access

Law is a formal system, but it is not a closed one. It depends on the notice, comprehension, and executive capacity of the humans it binds, and when those fail, the formal system fails with them.

This chapter supplies the jurisdictional and due-process grounding for the ADR–DYREO–PSDLEF architecture as a whole. Where Parts II and VIII addressed constitutional standing and the juridical form of the trust, this chapter states the specific statutory hooks on which the Maryland action turns, the *Mullane* notice problem created by DTC book-entry masking during the 2014 USEC/Centrus reorganisation, and the ADA Title II accommodation argument that connects the Gov-LLM protocol to the constitutional right of access to court.

Register

The statutes and cases cited are **[ADOPTED LAW]** law. The proposition that they compel the relief described is **[PROPOSAL]** framed by the petitioner, not a ruling of any court. The endpoint `https://americana-digital-reserve-exchange.ai.studio/` is a **[DESIGN TARGET]** cited as designed, not as a production deployment.

42.1 Equitable jurisdiction and the Maryland mandamus vehicle

The state-court petition in Circuit Court for Baltimore City, Case No. 24-C-26-003970, invokes the general equity jurisdiction conferred by Md. Code Ann., Cts. & Jud. Proc. § 1-501 (*Md. Code Ann., Cts. & Jud. Proc. § 1-501* n.d.), under which each circuit court sits as a court of general common-law and equity jurisdiction. The extraordinary-writ vehicle is Maryland Rule 15-701 (*Maryland Rule 15-701* n.d.), which governs mandamus and other writs issued in the original jurisdiction of the circuit court.

Two features of that vehicle are doing work. First, mandamus reaches a *ministerial* act: a duty so clearly owed that the respondent has no discretion to refuse it. The petition frames the duty as ministerial turnover of book-entry positions by the corporate transfer agent and by any successor entity holding the underlying record position. Second, specific performance in equity is the appropriate remedy where the res is unique, as a beneficial interest traceable through a Chapter 11 reclassification is, because damages do not put the holder back in the position the reclassification would have occupied had notice issued. Whether either characterisation is correct on the facts is for the Maryland court, and the chapter records, rather than resolves, the characterisation.

42.2 The *Mullane* due-process claim

Under the Fifth Amendment (as to federal proceedings) and the Fourteenth Amendment (as to state proceedings), and under *Mullane v. Central Hanover Bank & Trust Co.*, 339 U.S. 306 (1950) (Central Hanover Bank & Trust Co. 1950), deprivation of property requires notice "reasonably calculated, under all the circumstances, to apprise interested parties of the pendency of the action and afford them an opportunity to present their objections." *Mullane* holds that where the identities and addresses of beneficiaries are *reasonably ascertainable*, publication notice is constitutionally insufficient; individual notice is required.

The petition's constitutional claim is that multi-tiered Depository Trust Company (DTC) book-entry masking during the 2014 Chapter 11 reorganisation of USEC Inc. into Centrus Energy Corp. (Case No. 14-10492, Bankr. D. Del.) (*USEC Inc., Case No. 14-10492 (Bankr. D. Del.)* 2014) produced precisely the *Mullane* defect: beneficial holders whose identities were ascertainable with reasonable diligence through SEC Form 13F institutional-holder filings (Securities Exchange Act of 1934 1934) were not given individual notice of the reclassification, with the result that their positions were cancelled in the reorganisation without the opportunity to be heard.

Register

That the reorganisation plan confirmed in 2014 produced the emergence of Centrus Energy Corp. and a 0.0917:1 reverse split is a matter of public corporate-action record (*Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history* 2014; *USEC Inc., Case No. 14-10492 (Bankr. D. Del.)* 2014) and is **[ADOPTED LAW]** for purposes of this chapter. That DTC-tier masking caused a *Mullane* violation as to any particular holder is **[PROPOSAL]** pleaded in the petition, not an adjudication.

The doctrinal significance of the claim is that it reframes what might otherwise be treated as a stale securities-law problem as a constitutional-notice problem to which no ordinary limitation period cleanly attaches where the deprivation is ongoing through the current continued refusal of transfer agents to honour the reclassified positions.

42.3 ADA Title II cognitive accommodations and the Gov-LLM endpoint

The second constitutional hook is access to court. Under *Tennessee v. Lane*, 541 U.S. 509 (2004) (Lane 2004), Congress validly abrogated state sovereign immunity under Title II of the ADA with respect to the fundamental right of access to the courts. Under 42 U.S.C. § 12131 (Americans with Disabilities Act of 1990 1990) and 28 C.F.R. § 35.130(b)(7) (*28 C.F.R. § 35.130(b)(7)* n.d.), a public entity must make reasonable modifications in policies, practices, or procedures when necessary to avoid discrimination on the basis of disability.

The petition argues that a *pro se* neurodivergent litigant facing a ministerial turnover order, a Chapter 11 plan, a DTC book-entry chain, and a Tucker Act compensation proceeding simultaneously is, without a cognitive accommodation, structurally excluded from vindicating the very rights *Lane* protects. The accommodation requested is integration of the Gov-LLM Secure Transaction Protocol and automated API transfer endpoints (`https://americana-digital-reserve-exchange.ai.studio/`) into the filing and transfer workflow, so that the execution steps the law requires are reduced to a cryptographically auditable sequence rather than left to the unassisted executive capacity of the litigant.

This is the juncture at which this book's thesis about cognitive infrastructure (Parts IV, X) meets its thesis about access to justice (Parts II, IX): an accommodation that is itself software, rather than a physical ramp or an interpreter, is an unusual request but a logically compelled one if the disability in question is executive dysfunction rather than mobility or sensory impairment.

CHAPTER 43

Capital Restructuring Taxonomy and CUSIP Normalization

Every securities claim is, in the first instance, an identifier claim. The rest is arithmetic.

This chapter states the capital-structure facts that the petition rests on, runs the arithmetic from the 1999 indenture through the 2014 plan, and shows how CUSIP check-digit validation is used to reconstruct degraded institutional-holder records. Chapter 27 (Part VIII) described the CUSIP normalisation protocol procedurally; this chapter gives the underlying capital vector and the precise numbers it produces.

Register

The indenture principal amounts, coupons, maturity dates, and CUSIPs reported in Sections 2.1–2.2 are taken from the petition and are recorded here on the petitioner's representation; I have not independently pulled the 1999 indenture from EDGAR and mark these figures **[CITE REQUIRED]**. The 0.0917 plan exchange ratio is confirmed against public corporate-action records (*Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history* 2014). The consolidated $720,150,000 total capitalisation vector is **[PROPOSAL]**, because it depends on (a) Tucker Act and patent allocation components that are themselves claims, not judgments.

43.1 The 1999 legacy indenture

On 14 January 1999, USEC Inc. issued $500,000,000 in Senior Notes bifurcated into two series:

Table 43.1: USEC Inc. Senior Notes issued 14 January 1999.

Series	Principal	Coupon	Maturity	CUSIP
Series A	$350,000,000	$6\frac{5}{8}\,\%$	2006	917300AA1
Series B	$150,000,000	$6\frac{3}{4}\,\%$	2009	917300AB9
Total	$500,000,000			

The two CUSIPs follow the standard nine-character format: the first six characters (917300) identify USEC Inc. as issuer, the seventh and eighth (AA, AB) identify the two series, and the ninth (1, 9) is the check digit. Validation against the modified Luhn algorithm is what permits ingestion pipelines to reject transcription errors before they propagate into the entitlement calculation.

43.2 The 2014 Chapter 11 reclassification

Pursuant to the confirmed plan of reorganisation in Case No. 14-10492 (*USEC Inc., Case No. 14-10492 (Bankr. D. Del.)* 2014), the Senior Note claims were reclassified as Class 4 Allowed Unsecured Claims exchangeable into reorganized common stock of the emerging entity, Centrus Energy Corp. (NYSE: LEU), at the court-confirmed ratio of 0.0917 shares of LEU common per dollar of allowed claim. The ratio matches the 0.0917:1 reverse split recorded in the public corporate-action records for the effective date of 30 September 2014 (*Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history* 2014).

43.3 The consolidated capital vector

Multiplying the \$500,000,000 principal of the legacy indenture by the confirmed exchange ratio of 0.0917 yields an exact entitlement of

$$500{,}000{,}000 \times 0.0917 \;=\; 45{,}850{,}000 \text{ shares of LEU common.} \tag{43.1}$$

Combining this reclassified equity position with the separate Tucker Act judgment vector and the patent-licensing allocation vector (the latter pursued under 28 U.S.C. § 1498(a) before the Court of Federal Claims, as discussed in Chapter 27) produces the consolidated capitalisation figure the petition asserts as the fully assembled endowment:

$$V_{\text{total}} \;=\; \$720{,}150{,}000.00. \tag{43.2}$$

Register

Equation (43.2) is **[PROPOSAL]**. Its two components other than the \$500M × 0.0917 legacy tranche are asserted claims, not confirmed amounts. The figure is reproduced here because it is the number the quantitative engine in Chapter 43 takes as its starting corpus, not because the book represents that it has been awarded.

43.4 CUSIP check-digit validation against Form 13F filings

Automated ingestion pipelines resolve historical data degradation via modified Luhn check-digit validation run across institutional Form 13F filings, including PNC Financial Services Group (SEC File No. 28-1235) and other named holders. The routine is: convert alphabetic CUSIP characters to their ordinal values (A=10, B=11, …, Z=35) with the convention documented by the CUSIP Global Services standard (CUSIP Global Services 2020); weight positions from right to left, doubling at alternating positions; sum the digits of the resulting products; the check digit is $(10 - (P \bmod 10)) \bmod 10$. Any identifier that fails this check is rejected before it touches the entitlement ledger, which prevents the kind of transcription corruption that otherwise masks the chain of title across a Chapter 11 reorganisation.

This is the same procedure described in Chapter 27, but applied here to a specific indenture rather than stated in the abstract. It is deliberately boring: registry mechanics are unglamorous and decisive.

CHAPTER 44

DYREO Stochastic Control and the 31-Algorithm Suite

Writing down an objective function is a normative act disguised as a mathematical one. The previous chapter said so at the level of a one-period quadratic program; this chapter says so at the level of continuous-time stochastic control.

Chapter 25 presented DYREO in its one-period mean-variance form. This chapter elevates it to the continuous-time Hamilton–Jacobi–Bellman (HJB) form that the Ph.D.-level synthesis specifies, states the closed-form daily yield-sweep vector, and gives the Social-Energy-Return-on-Investment (S-EROI) metric in the thermodynamic form used by the solar/battery pilot.

44.1 Continuous-time stochastic control

The Dynamic Yield & Real Estate Optimization (DYREO) algorithm models liquid trust assets in continuous time as a controlled diffusion:

$$dA(t) \;=\; \mu A(t)\,dt \;+\; \sigma A(t)\,dW(t) \;-\; u(t)\,dt, \tag{44.1}$$

where $A(t)$ is liquid trust assets at time t, μ is the drift, σ is the volatility, $W(t)$ is a standard Brownian motion, and $u(t)$ is the distributed-yield control (the rate at which the trust disburses to its four daily buckets). The trust solves

$$J(t,a) \;=\; \sup_{u(\cdot)\in\mathcal{U}} \mathbb{E}\left[\int_t^T e^{-\rho(s-t)} U\big(A(s),u(s),S(s)\big)\,ds \;\Big|\; A(t)=a\right], \tag{44.2}$$

where U combines financial return, social-utility return from the Baltimore solar microgrid, distribution to ADA *pro se* AI-grants, and operational reserves, subject to the capital-preservation constraint

$$\Pr\big(A(T)\ge (1-\delta)A(0)\big) \;\ge\; 1-\epsilon, \tag{44.3}$$

and to the deterministic yield-sweep vector of Section 3.2. The HJB equation is the standard one for problems of this class (Bellman 1957; Björk 2020):

$$-\frac{\partial J}{\partial t} \;=\; \sup_{u}\Big\{U(a,u,s) + (\mu a - u)\frac{\partial J}{\partial a} + \tfrac{1}{2}\sigma^2 a^2 \frac{\partial^2 J}{\partial a^2} - \rho J\Big\}. \tag{44.4}$$

Register

Equations (44.1)–(44.4) are standard continuous-time portfolio-control machinery and are **[ADOPTED LAW]** in form. The calibration of μ, σ, ρ, δ, and ϵ to the PSDLEF corpus is **[DESIGN TARGET]**, as are the resulting optimal $u(t)$ trajectories.

44.2 The closed-form daily yield-sweep matrix

Liquid proceeds are distributed each business day by a deterministic weight vector that fixes, by policy, how incoming yield is swept into the four operational buckets before any discretionary reinvestment occurs:

Table 44.1: Daily yield-sweep vector (closed-form policy).

Bucket	Weight	Purpose
Treasury Equivalents	60%	Capital-preservation and permanent-corpus floor
Baltimore Solar Microgrid Payables	25%	Pilot service, low-income housing credits, maintenance
ADA *Pro Se* Litigant AI Grants	10%	Gov-LLM access grants under the Title II accommodation theory
Operational Reserves	5%	Trust operations, audit, custody
Total	100%	

The 60/25/10/5 split is a **[PROPOSAL]** policy choice, not a solution to the HJB equation (44.4); it is the floor the HJB optimisation must respect rather than an output of it. That is the point of the design: the social-distribution buckets are protected from being optimised away by a pure-financial objective function, because they are imposed as a deterministic sweep before optimisation runs.

44.3 The Social Energy Return on Investment (S-EROI)

Real-estate and microgrid allocations are governed by a thermodynamic efficiency metric that compares lifetime delivered energy net of losses to lifetime capital and maintenance cost:

$$\text{S-EROI} \;=\; \frac{\int_0^T \left[P_{\text{solar}}(t) - P_{\text{loss}}(t)\right] dt}{K_{\text{cap}} + \int_0^T C_{\text{maint}}(t)\, dt} \;\geq\; 0.73 \text{ kWh/\$}. \tag{44.5}$$

The 0.73 kWh/$ threshold is the design target against which each rooftop installation is accepted or rejected in the pilot. It is tighter than the S-EROI reported in Chapter 25 because the version here is calibrated for low-income housing, where the marginal social value of reliable delivered kilowatt-hours is higher.

Register

The 0.73 kWh/$ threshold and the solar-pilot parameters underlying P_{solar}, P_{loss}, K_{cap}, and C_{maint} are **[DESIGN TARGET]**.

44.4 The 31-algorithm execution suite

The executive financial engine spans thirty-one algorithms across the modules listed in Chapter 25, covering:

- *Bond valuation* (Algorithms 01–03): discounted cash-flow pricing, yield-to-maturity, and credit-spread term structure for the legacy indenture and any newly issued trust certificates.
- *Asset conversion and distribution* (04–06): CUSIP normalisation (Chapter 42), plan-ratio application, yield-sweep execution against Table 44.1.

- *Volatility modelling* (07–09): GARCH(1,1) estimation of return volatility (Bollerslev 1986), EWMA variance, and jump-diffusion calibration for extreme-event days.
- *Execution optimisation* (10–12): optimal trade-slicing, market-impact modelling, and rebalancing under transaction costs.
- *Derivatives and hedging* (13–15): Black–Scholes option pricing (Black and Scholes 1973), Greek computation, tail-risk collar construction on the equity tranche.
- *Risk management* (16–18): Value-at-Risk (VaR), Conditional VaR (CVaR/Expected Shortfall), and stress-testing against the capital-preservation constraint (44.3).
- *Yield optimisation* (19–20): HJB solution methods for (44.4), including numerical finite-difference solvers and verification against closed-form special cases.
- *Automated market-maker liquidity* (21–22): constant- product and concentrated-liquidity AMM curves for any tokenised beneficial-interest positions the trust issues.
- *Hash-chain audit trail and compliance* (23–31): the immutable transaction log described in Chapter 44 and Part VI, with ML-DSA-87 signatures on every ledger entry.

The suite is a **[PROPOSAL]**: the algorithms are individually standard, but their composition into a single auditable pipeline is the design the book describes.

CHAPTER 45

Post-Quantum Enclaves, QRDE-AEAD, and the DAG/Diamond Pipeline

A harvest-now, decrypt-later adversary does not care that you will upgrade your cryptography in 2035. They are recording your ciphertext today.

Chapter 15 (Part IV) and Chapter 20 (Part VI) introduced the post-quantum security module and the RIGOR-PRS-Secure++ scheme. This chapter pins down the specific NIST FIPS 203/204 parameter choices, the dual-envelope QRDE-AEAD construction, the deterministic DAG execution model over MMIO registers, and the integration of diamond color-center qubits as first-class DAG task nodes.

45.1 NIST FIPS 203 and FIPS 204 enclaves

To neutralise "harvest-now, decrypt-later" attacks, all data enclaves and custodial transactions use the NIST-standardised lattice primitives at the highest parameter sets:

- **ML-KEM-1024** (Module-Lattice-Based Key Encapsulation Mechanism, FIPS 203) (National Institute of Standards and Technology 2024c) for asymmetric key establishment, providing NIST security level 5 (equivalent to AES-256 post-quantum strength).
- **ML-DSA-87** (Module-Lattice-Based Digital Signature Algorithm, FIPS 204) (National Institute of Standards and Technology 2024b) for all transaction signatures, audit-log entries, and trust distributions.
- **SLH-DSA** (Stateless Hash-Based Digital Signature, FIPS 205) (*FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA)* 2024) as a conservative fallback for long-lived root-of-trust signatures, because hash-based signatures retain security under conservative cryptographic assumptions.

Register

The standards are **[ADOPTED LAW]** (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; *FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA)* 2024). The statement that the PSDLEF deployment uses ML-KEM-1024 and ML-DSA-87 is a **[DESIGN TARGET]**: these are the parameter choices the specification mandates, not a deployed configuration.

45.2 The QRDE-AEAD dual-envelope protocol

The Quantum-Resistant Dual-Envelope Authenticated Encryption with Associated Data construction combines asymmetric KEM encapsulation with symmetric authenticated encryption and an outer authentication tag to produce an IND-CCA2-secure hybrid:

1. Generate or reuse a long-lived ML-KEM-1024 encapsulation key pair for the enclave.
2. Encapsulate a fresh ephemeral shared secret ss via ML-KEM-1024 to produce the ciphertext c_K and shared secret K.
3. Derive two symmetric keys via HKDF-SHA3-384 (Krawczyk and Eronen 2010) from K: a data-encryption key k_e and an authentication key k_a.
4. Encrypt the payload m with AES-256-GCM (Dworkin 2007) under k_e to produce the inner ciphertext c_e and GCM tag t_g.
5. Compute an outer HMAC-SHA3-256 tag t_o over $(c_K \parallel c_e \parallel t_g \parallel ad)$ under k_a, where ad is the associated data (sender, timestamp, ledger height, CUSIP).
6. The transmitted envelope is (c_K, c_e, t_g, t_o, ad).

The dual-envelope structure ensures that the post-quantum asymmetric layer and the symmetric layer fail independently: a compromise of the KEM does not expose the payload without a break of AES-256-GCM, and a compromise of the symmetric layer is useless against a passive adversary without the long-lived KEM private key.

45.3 Deterministic DAG execution over MMIO

The execution engine replaces imperative blocking loops with a pure Directed Acyclic Graph (DAG) of topological transitions. Each node is a pure function over its fan-in edges; the scheduler executes a node exactly once, when every fan-in edge has delivered a valid token. This gives deterministic, replayable execution without global locks.

The hardware abstraction layer is mapped to:

- **MMIO register controllers** at base address `0x40000000`, with per-engine register windows for KEM acceleration, signature verification, hash, and DYREO control.
- **Fan-in gate counters** K_{in} enforced in hardware, so that a node fires if and only if exactly its declared number of predecessor tokens have arrived.
- **DWARF debug symbol offsets** for deterministic trace and post-hoc audit of every DAG firing, so that the execution trace can be reconstructed from the ledger even after the fact.

This is the same execution model described architecturally in Chapter 12 (Part IV); the DAG here is the one that the 31-algorithm suite actually runs on.

45.4 The LoongDiamond-2.5 color-center nodes

Chapter 13 introduced the LoongDiamond-2.5 quantum substrate. In the fully specified pipeline, diamond color-center qubits — nitrogen-vacancy (NV), silicon-vacancy (SiV), germanium-vacancy (GeV), and tin-vacancy (SnV) — are represented as first-class DAG task nodes. Spin coherence times, microwave control pulses, and photonic interconnect events map directly into tokens on the DAG edges, so that a quantum control step is a node like any other, with deterministic fan-in and fan-out from the perspective of the surrounding classical DAG.

Register

The existence, coherence, and control characteristics of color-center qubits are [**ADOPTED LAW**] in the condensed-matter literature. Integration of those qubits as first-class DAG nodes in the SSE-9/ZU9EG carrier (Chapter 14) is [**DESIGN TARGET**].

CHAPTER 46

SSADYREO, the MORTAR LSH Priority Score, and the OSI-P Audit Chain

A public-safety net does not fail all at once. It fails claim by claim, minute by minute, as decisions that should take days take years and people who cannot wait die waiting. A perpetual trust can capitalise the programme; an algorithmic triage engine can keep the queue from killing people while the capital works.

This chapter combines the perpetual-purpose trust structure of Chapter 24, the MORTAR triage engine of Chapter 17, and the OpenSSA integrity pipeline of Chapter 19 into the SSADYREO perpetual-trust stabilization architecture, and pins down the $O(n \log m)$ LSH priority scoring, the $N_{\text{sup}} = 16$ supervised transformer block configuration, and the nine-stage R_1–R_9 cryptographic audit chain.

46.1 Perpetual purpose trust under the UDTA

The Social Security Administration DYREO trust (SSADYREO) is modelled as a perpetual purpose trust under the Uniform Directed Trust Act (*Uniform Directed Trust Act* 2017) with a quad-cameral separation of fiduciary powers (Investment Committee, Distribution Committee, Trust Protector, Independent Examiner) that mirrors the four-component architecture introduced in Chapter 23. The central perpetuity constraint is a hard distribution cap:

$$d(t) \;\le\; 0.05 \times \overline{A}_{5\text{yr}}(t), \tag{46.1}$$

where $d(t)$ is annual distributions and $\overline{A}_{5\text{yr}}(t)$ is the trailing five-year moving average of trust corpus. This 5% rule permanently immunises public safety-net programmes from principal depletion, because the trust cannot distribute more than five per cent of its smoothed value in any year regardless of political or actuarial pressure.

Register

The UDTA as a model statute is **[ADOPTED LAW]** (*Uniform Directed Trust Act* 2017). The 5% trailing-average distribution rule of (46.1) is a **[PROPOSAL]** policy constraint (it is similar to but stricter than the 5% qualifying-distribution rule familiar from private-foundation tax law, and is imposed here by design rather than by any tax code).

46.2 MORTAR $O(n \log m)$ LSH priority scoring

The Mortality-Optimized Risk-Triage And Reasoning (MORTAR) engine is specified to eliminate administrative mortality across the approximately 1.1-million-claimant SSA adjudicative backlog using locality-sensitive hashing (LSH) priority scoring (Gionis et al. 1999b; Datar et al. 2004) in $O(n \log m)$ time, where n is the number of incoming claims and m is the number of buckets in the hash table. The priority score for a claim vector x is

$$s(x) \;=\; w_{\text{mort}} \cdot p_{\text{mort}}(x) \;+\; w_{\text{sev}} \cdot p_{\text{sev}}(x) \;-\; w_{\text{wait}} \cdot \tau(x), \tag{46.2}$$

where p_{mort} is the predicted short-term mortality probability, p_{sev} is severity, τ is current wait time, and the w are calibrated weights. Claims are hashed into m buckets via k random hyperplanes; candidates within a bucket are scored exactly, giving sublinear retrieval of the highest-priority claimants.

The scoring function feeds iterative latent-reasoning transformer blocks with $N_{\text{sup}} = 16$ supervised layers, which produce the structured recommendation that the human adjudicator then decides. The adjudicator decides; the engine prioritises and drafts. That division of labour is the non-delegation guard-rail: the engine does not adjudicate, because adjudication is a governmental function that a private algorithm may not exercise.

46.3 The OpenSSA Integrity Pipeline (OSI-P): R_1–R_9

The OpenSSA Integrity Pipeline enforces a nine-stage cryptographic audit chain anchored to Merkle roots (Merkle 1979; Merkle 1980) and signed via FIPS 204 ML-DSA-87 at every stage:

Table 46.1: The nine-stage OSI-P audit chain.

Stage	Name	Function
R_1	Ingest	Tamper-evident receipt of incoming claim record, ML-DSA-87 signed by submitter.
R_2	Normalise	Canonical JSON-LD normalisation and CUSIP/identifier validation.
R_3	Hash	Per-record SHA3-512 digest appended to the stage Merkle tree.
R_4	Score	MORTAR priority score (46.2) recorded with feature hash.
R_5	Recommend	Transformer-block recommendation recorded with model-version hash.
R_6	Adjudicate	Human-adjudicator decision recorded with adjudicator ML-DSA-87 signature.
R_7	Distribute	Distribution execution recorded under the DYREO sweep (Table 44.1).
R_8	Reconcile	Daily cross-ledger reconciliation; root of roots written to the trust anchor.
R_9	Seal	Day's Merkle root signed with long-lived SLH-DSA root key for archival permanence.

The result is a self-authenticating electronic administrative record admissible under Federal Rules of Evidence 902(13)–(14) (*Federal Rule of Evidence 902(13)–(14)* n.d.): a certified record generated by an electronic process or system that produces an accurate result, with every stage cryptographically tied to the next and no undocumented write path.

46.4 How the three layers compose

The composition is the point of the synthesis, and it is worth stating in one place:

1. The UDTA purpose trust of Section 5.1 provides the legal permanence and the 5% distribution floor.
2. The DYREO stochastic-control engine of Chapter 43, running on the DAG substrate of Chapter 44, allocates capital within that floor and executes the daily yield sweep.
3. The MORTAR triage engine of Section 5.2 prioritises the 1.1-million-claim queue without adjudicating it.
4. The OSI-P nine-stage chain of Section 5.3 produces an evidentiary record that is admissible in federal court without extrinsic authentication testimony.

5. The QRDE-AEAD dual-envelope encryption of Section 4.2 protects every stage from quantum-capable adversaries.

None of the five layers is sufficient alone. A perpetual trust without an allocation engine is an inert corpus. An allocation engine without a triage layer optimises yield while people die. A triage layer without an integrity pipeline produces inadmissible — and therefore useless — records. A record chain without post-quantum cryptography is a record chain built for a decade that will not survive the adversary it is designed to face.

CHAPTER A

Filing-Readiness Checklist

This checklist is the one the working draft of the statement of facts applies to itself. It is reproduced here because it states, in operational form, the discipline that Parts IX and XI describe. An item that cannot be checked is a proposition that cannot yet be filed.

1. Confirm the court, the caption, the docket number, and the purpose of the statement.
2. Confirm whether the document is a brief, a petition, a motion, a declaration, or a memorandum, and apply the page limits and rules that govern that form.
3. Obtain the official docket sheet for every referenced proceeding.
4. Reconcile the state case-number discrepancy between the two Maryland case numbers carried in the source materials (D-01-CR-25-011312 and D-01-CR-25-011146), determining which number belongs to which proceeding. **[CITE REQUIRED]**
5. Confirm the exact disposition entered on 30 January 2026 and its scope. **[CITE REQUIRED]**
6. Obtain file-stamped versions of every complaint, petition, and motion referred to.
7. Verify every appellate and Supreme Court docket number.
8. Obtain every order, judgment, and opinion cited.
9. Identify the exact Social Security decision under review and the certified administrative record.
10. Confirm which scientific and genomic exhibits were submitted and included in the record.
11. Locate every accommodation request, the response to it, and proof of receipt.
12. Authenticate each technical report, and label each system as proposed, simulated, tested, or deployed.
13. Do not describe an HMAC-SHA3-256 construction as ML-DSA, and do not describe simulated output as validated cryptography (National Institute of Standards and Technology 2024b).
14. Remove any unsupported claim of clinical validation, certification, government deployment, or legal admissibility.
15. Attribute each claim and each request for relief to the correct proceeding.
16. Review sealing, redaction, privacy, and protective-order requirements.
17. Replace every citation-required marker with a citation or remove the proposition the marker supports.
18. Confirm that each citation supports the exact proposition for which it is offered.
19. Reconcile the chronology with the operative pleading and the relief sought in the current case.
20. Review compliance with the local rules of the relevant court and with the applicable appellate rules.

CHAPTER B

Citation Templates

The templates below are forms, not sources. Each placeholder must be replaced with verified information before the template is used. They are collected here because a party proceeding without counsel is most often defeated not by the strength of the opposing argument but by the form of the citation.

Federal docket entry.
[Caption], No. [number], ECF No. [number], at [page] ([court] [date]).

Order.
Order, *[Caption]*, No. [number], at [page] ([court] [date]).

Transcript.
Hr'g Tr. [date], at [page]:[line]–[line].

Administrative record.
Administrative Record (AR) [page], [document title and date].

State docket.
Maryland Judiciary Case Search, [court], Case No. [verified number], docket entry dated [date], [entry title].

Technical report.
[author or entity], *[title]*, version [version], [date], at [page or section], [stable identifier].

A citation is not a decoration. It is a promise that the document exists, that it says what the sentence says, and that it can be produced on request. A citation that cannot be verified is worse than no citation, because it converts a gap in the record into an appearance of support.

CHAPTER C

Recommended Record Assembly Order

The order in which a record is assembled determines whether a reviewer can follow the argument. The order below proceeds from the procedural skeleton inward to the substantive exhibits, and it ends with the table that ties each paragraph of the statement to its source.

1. A master docket index that lists every case separately.
2. The official docket sheet and the operative pleading for each case.
3. Orders and judgments in chronological order.
4. Notices of appeal, petitions, and appellate dispositions.
5. The Social Security administrative decision, the notices, and the certified record.
6. The exhibits that were submitted, each with proof of submission.
7. Accommodation requests, responses, and transcripts.
8. The scientific reports and the source publications relied upon.
9. Source-code snapshots, test outputs, and validation reports.
10. A paragraph-to-source citation table for the completed statement.

CHAPTER D

Source-File Index

This book was assembled from a project archive and from two consolidated documents. The index below records the principal sources so that a reader can trace any chapter to the material it was built from. The archive documents are working papers, not publications; they are cited in this book only as the origin of a claim, and the discipline of Chapter 2 governs their use.

D.1 The consolidated source documents

PSDLEF Sovereign Exchange Trust: Integrated Master Architecture.
156 pages, October 2026. Contains the HSPA v3.0 / SSE-9 / RIGOR-PRS master technical specification, the ten-volume master portfolio, the integrated master architecture, and appendices on filing readiness, citation templates, and record assembly. The base for Parts I–VIII and for Appendices A–C.

Final Consolidated Master Academic, Technical, & Jurisprudential Portfolio.
The ten-volume framework for degree conferral, human-rights advocacy, and trust infrastructure. Source of the volume structure that Parts I–XI follow.

Integrated Framework for Cognitive Accessibility, Genomic Justice, and Quantum-Secure Infrastructure.
Document reference UN/PSDLEF/GFQA/2026/001. Source of the three-pillar directive in Part X.

D.2 The archive documents

Section 1983 civil-rights analysis.
The five counts, their elements, and their doctrinal support. Base for Part II.

Master consolidated petition, McLaughlin v. Maryland.
The civil-rights pleading working draft. Base for Part II and Part IX.

CAFC consolidated petition.
The Federal Circuit materials, including the patent and sovereign-immunity arguments. Base for Part II and Part IV.

Master portfolio: genomic privacy, five volumes.
The genomic-privacy architecture, its constitutional argument, and its statutory analysis. Base for Part II.

Expanded neurophysiological report.
The GWAS, polygenic-score, and neuroimaging materials. Base for Part III.

State-government LLM legal framework.
The statutory lattice, the knowledge base design, and the compliance framework. Base for Part X.

UMGC integrated capstone manuscript.
The closed-loop neuroprosthetic design and its validation plan. Base for Part XI.

United Nations presentation.
The mortality and backlog record and the treaty framework. Base for Part VII and Part XI.

Chapter 11 plan of reorganisation.
The classification of claims and the securities liquidation. Base for Part XI and Part VIII.

Patent application materials.
The clean-slate claim set and its *Alice* exposure. Base for Part IV.

Accessibility materials.
The cognitive-accessibility requirements. Base for Part X.

Legal-compliance framework and executive summaries.
The integration reports that connect the several projects. Base for Part I.

Status of the archive

The archive documents are **[PROPOSAL]** and **[MODEL / SIMULATION]** material. Nothing in them is adopted law merely because it appears in the archive, and nothing in this book should be cited to the archive as if the archive were an authority. The archive's function is to be the record of what the author proposed; the authority for a legal proposition is the statute, rule, or decision, and nothing else.

Table of Authorities

The authorities cited in this volume are collected here in two groups. The first lists the reported decisions and the reasoning each supplies. The second lists the constitutional and statutory provisions, the treaties, the standards, and the executive and legislative instruments. Full bibliographic details appear in the bibliography.

Cases

***Alice Corp. v. CLS Bank Int'l*, 573 U.S. 208 (2014).**
Patent eligibility of claims drawn to abstract ideas.

***Board of Regents v. Roth*, 408 U.S. 564 (1972).**
Property and liberty interests protected by procedural due process.

***Buckley v. Fitzsimmons*, 509 U.S. 259 (1993).**
The line between a prosecutor's investigative and advocative functions.

***Carpenter v. United States*, 585 U.S. 296 (2018).**
Fourth Amendment protection of digital records of a person's movements.

***City of Canton v. Harris*, 489 U.S. 378 (1989).**
Deliberate indifference and the failure-to-train theory of municipal liability.

***Cohen v. Beneficial Industrial Loan Corp.*, 337 U.S. 541 (1949).**
The collateral-order doctrine and appealable interlocutory orders.

***Connick v. Thompson*, 563 U.S. 51 (2011).**
The single-incident limit on municipal liability.

***Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993).**
The trial judge's gatekeeping duty for scientific testimony.

***Ex parte Young*, 209 U.S. 123 (1908).**
Prospective injunctive relief against state officers notwithstanding sovereign immunity.

***Goldberg v. Kelly*, 397 U.S. 254 (1970).**
Pre-termination evidentiary hearings for welfare benefits.

***Gonzalez v. Trevino*, 602 U.S. 653 (2024).**
Narrow construction of the retaliatory-arrest exception.

***Harlow v. Fitzgerald*, 457 U.S. 800 (1982).**
The objective qualified-immunity standard.

***Heck v. Humphrey*, 512 U.S. 477 (1994).**
The favourable-termination requirement and the bar on claims that impugn an outstanding conviction.

***Imbler v. Pachtman*, 424 U.S. 409 (1976).**
Absolute prosecutorial immunity for the advocative function.

***Kumho Tire Co. v. Carmichael*, 526 U.S. 137 (1999).**
Extension of *Daubert* to non-scientific expert testimony.

***Manuel v. City of Joliet*, 580 U.S. 357 (2017).**
Pretrial detention without probable cause as a Fourth Amendment claim.

***Maryland v. King*, 569 U.S. 435 (2013).**
DNA identification and the Fourth Amendment.

***Mathews v. Eldridge*, 424 U.S. 319 (1976).**
The three-factor procedural-due-process balance.

***Monell v. Department of Social Services*, 436 U.S. 658 (1978).**
Municipal liability under 42 U.S.C. § 1983.

***Nieves v. Bartlett*, 587 U.S. 391 (2019).**
Retaliatory arrest requires motive plus the absence of probable cause.

***Olmstead v. L.C.*, 527 U.S. 581 (1999).**
The integration mandate of ADA Title II.

***Riley v. California*, 573 U.S. 373 (2014).**
The warrant requirement for searches of cell phones.

***Thompson v. Clark*, 596 U.S. 36 (2022).**
Favourable termination requires only a termination without conviction.

***United States v. Little Lake Misere Land Co.*, 412 U.S. 580 (1973).**
The Supremacy Clause and federal interests.

***Van de Kamp v. Goldstein*, 555 U.S. 335 (2009).**
Absolute immunity for administrative acts connected to the advocative function.

***Winter v. Natural Resources Defense Council*, 555 U.S. 7 (2008).**
The four-factor preliminary-injunction standard.

***Younger v. Harris*, 401 U.S. 37 (1971).**
Abstention from pending state criminal proceedings.

Constitutions, statutes, treaties, and standards

U.S. Const. amends. IV, V, XIV.
The constitutional provisions on which Parts II and X rest.

42 U.S.C. § 1983.
Civil action for deprivation of rights.

28 U.S.C. § 1291.
Final decisions of district courts.

42 U.S.C. § 405(g).
Judicial review of Social Security decisions.

28 U.S.C. § § 1491, 1498, 1631.
Tucker Act jurisdiction, patent claims against the United States, and transfer to cure a jurisdictional defect.

Americans with Disabilities Act of 1990; ADA Amendments Act of 2008.
Nondiscrimination and accommodation.

29 U.S.C. § 794.
Section 504 of the Rehabilitation Act.

20 U.S.C. § § 1400, 1232g.
IDEA and FERPA.

42 U.S.C. § 2000ff.
The Genetic Information Nondiscrimination Act.

42 U.S.C. § 1320d.
HIPAA and the Security Rule.

5 U.S.C. § 551.
The Administrative Procedure Act.

35 U.S.C. § § 200–212.
The Bayh–Dole Act.

11 U.S.C. § 101 *et seq.*; Subchapter V.
The Bankruptcy Code and the Small Business Reorganization Act of 2019.

International Covenant on Civil and Political Rights (1966).
Civil and political rights, including the non-derogable right to life.

Convention against Torture (1984).
Prohibition of cruel, inhuman, or degrading treatment.

Convention on the Rights of Persons with Disabilities (2006).
Accessibility, equal recognition, independent living, and habilitation.

FIPS 202, 203, 204, 205; FIPS 140-3.
SHA-3 and the three post-quantum standards, with the module-security requirements.

NIST SP 800-207.
Zero Trust Architecture.

NIST AI 100-1; NIST AI 600-1.
The AI Risk Management Framework and its generative-AI profile.

OMB Memorandum M-24-10.
Federal agency governance and risk management for AI.

E.O. 14110; E.O. 14179; *Winning the Race: America's AI Action Plan.*
The federal AI trajectory.

California SB 53, SB 813, AB 1405; Colorado SB 24-205.
State algorithmic- governance instruments.

NSA CNSA 2.0.
The federal post-quantum migration schedule.

Glossary of Terms and Acronyms

ADAAA
ADA Amendments Act of 2008.

ADMT
Automated decision-making technology.

APA
Administrative Procedure Act.

BSBS
Baltimore Secure Backbone System, the three-tier zero-trust municipal network design of Part XI.

CINP
Closed-loop neuroprosthetic, the capstone system of Part XI.

CNSA 2.0
Commercial National Security Algorithm Suite 2.0, the federal post-quantum migration schedule.

CRPD
Convention on the Rights of Persons with Disabilities.

CUSIP
Committee on Uniform Security Identification Procedures; the nine-character identifier for a security.

DYREO
The constrained portfolio-optimisation formulation of Part VIII, balancing return, social impact, and capital preservation.

GINA
Genetic Information Nondiscrimination Act of 2008.

Gov-LLM
The state-government legal-knowledge system proposed in Part X.

GPRS-Secure++
The post-quantum protection profile of the three-pillar directive.

HARMONI-X
The proposed cognitive-accessibility assistant.

HIMSS
Healthcare Information and Management Systems Society; its maturity model is used in the source materials as an architectural target only.

HSPA v3.0
Hybrid Semantic Processing Architecture, version 3.0, the clean-slate processor design of Part IV.

ICCPR
International Covenant on Civil and Political Rights.

LSH
Locality-sensitive hashing, the technique underlying the MORTAR matching engine.

ML-DSA, ML-KEM, SLH-DSA
The three post-quantum standards published in FIPS 204, 203, and 205 respectively.

MORTAR
Mortality-Optimised Risk-Triage engine, the bounded-delay adjudication queue of Part V.

NV centre
Nitrogen-vacancy centre in diamond, the spin defect used as the quantum substrate of Part IV.

OSI-P
OpenSSA Integrity Pipeline, the nine-stage provenance pipeline of Part VI.

PMCU
Probabilistic micro-controller unit, the scheduling layer of the HSPA control plane.

PQC
Post-quantum cryptography.

PRS
Polygenic risk score.

PSDLEF
Pro Se Disability Litigant Empowerment Foundation.

RLC
Reconfigurable logic cell, the morphable compute fabric of HSPA.

SEG
Semantic execution graph, the directed acyclic graph of semantic nodes.

SSE-9
Stochastic Semantic Engine, the asymmetric refinement processor of Part IV.

SSA, SSDI, SSI
Social Security Administration; Social Security Disability Insurance; Supplemental Security Income.

ZU9EG
The Zynq UltraScale+ device used as the carrier for the SSE-9 processor.

Bibliography

28 C.F.R. § 35.130(b)(7) (n.d.). Code of Federal Regulations. A public entity shall make reasonable modifications in policies, practices, or procedures when necessary to avoid discrimination on the basis of disability.

28 U.S.C. 1291 (2018). *Final Decisions of District Courts*. United States Code.

28 U.S.C. 1491 (2018). *Claims Against the United States (Tucker Act)*. United States Code.

29 U.S.C. 794 (2018). *Nondiscrimination Under Federal Grants and Programs (Section 504 of the Rehabilitation Act of 1973)*. United States Code.

42 U.S.C. 1983 (2018). *Civil Action for Deprivation of Rights*. United States Code.

42 U.S.C. 405(g) (2018). *Judicial Review of Decisions of the Commissioner of Social Security*. United States Code.

ADA Amendments Act of 2008 (2008). *Public Law 110–325, 42 U.S.C. 12101 note*.

Administrative Procedure Act (1946). *5 U.S.C. 551 et seq.* United States Code.

Alice Corporation v. CLS Bank International (2014). *573 U.S. 208*. Patent eligibility of abstract ideas.

Americans with Disabilities Act of 1990 (1990). *Public Law 101–336, 42 U.S.C. 12101 et seq.*

Bankruptcy Code (1978). *11 U.S.C. 101 et seq.* United States Code.

Bayh–Dole Act (1980). *35 U.S.C. 200–212 (Patent Rights in Inventions Made with Federal Assistance)*. United States Code.

Bellman, R. (1957). "Dynamic Programming and Markov Decision Processes". In: *Journal of Mathematics and Mechanics* 6.5, pp. 679–684.

Björk, T. (2020). *Arbitrage Theory in Continuous Time*. 4th. Oxford University Press.

Black, F. and M. Scholes (1973). *The Pricing of Options and Corporate Liabilities*. Vol. 81. 3, pp. 637–654.

Bollerslev, T. (1986). "Generalized Autoregressive Conditional Heteroskedasticity". In: *Journal of Econometrics* 31.3, pp. 307–327.

Buckley v. Fitzsimmons (1993). *509 U.S. 259*. Investigative versus advocative function.

California State Legislature (2025). *Senate Bill 53, Transparency in Frontier Artificial Intelligence Act*.

— (2026a). *Assembly Bill 1405, Artificial Intelligence Auditor Registry*.

— (2026b). *Senate Bill 813, Independent Verification Organizations for Artificial Intelligence Systems*.

Carpenter v. United States (2018). *585 U.S. 296*. Fourth Amendment and digital records of physical movement.

Central Hanover Bank & Trust Co., M. v. (1950). "339 U.S. 306". In: *United States Reports* 339. Due process requires notice reasonably calculated to apprise interested parties of the pendency of an action; publication suffices only for unknown or unascertainable parties., p. 306.

Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history (2014). Public corporate-action records. Confirmed reverse split of 0.0917:1 effective 30 September 2014 following emergence from Chapter 11.

City of Canton v. Harris (1989). *489 U.S. 378*. Deliberate indifference and failure to train.

Cohen v. Beneficial Industrial Loan Corp. (1949). *337 U.S. 541*. Collateral-order doctrine.

Colorado General Assembly (2024). *Senate Bill 24-205, Consumer Protections for Artificial Intelligence*. Replaced by SB 26-189, effective 1 January 2027.

Connick v. Thompson (2011). *563 U.S. 51*. Single-incident liability and prosecutorial training.

Convention against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment (1984). *1465 U.N.T.S. 85, entered into force 26 June 1987*.

Convention on the Rights of Persons with Disabilities (2006). *2515 U.N.T.S. 3, entered into force 3 May 2008*.

CUSIP Global Services (2020). *CUSIP Identifier Validation Guide*. S&P Global.

Datar, M., N. Immorlica, P. Indyk, and V. Mirrokni (2004). *Locality-Sensitive Hashing Scheme Based on p-Stable Distributions*. Proceedings of the 20th Annual Symposium on Computational Geometry (SoCG), 253–262.

Daubert v. Merrell Dow Pharmaceuticals (1993). *509 U.S. 579*. Trial-judge gatekeeping for expert scientific testimony.

Dworkin, M. J. (2007). *SP 800-38D: Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC*. National Institute of Standards and Technology.

Ex parte Young (1908). *209 U.S. 123*. Prospective injunctive relief against state officers.

Family Educational Rights and Privacy Act (2018). *20 U.S.C. 1232g*. United States Code.

Federal Rule of Evidence 702 (2023). *Testimony by Expert Witnesses*.

Federal Rule of Evidence 902(13)–(14) (n.d.). Federal Rules of Evidence. Self-authentication of certified records generated by an electronic process or process, and of certified data copied from an electronic device, storage medium, or file.

FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA) (2024). National Institute of Standards and Technology.

Gelber, A., T. Moore, Z. Pei, and A. Strand (2023). "Disability Insurance Income Saves Lives". In: *Journal of Political Economy* 131.11, pp. 3156–3185.

Genetic Information Nondiscrimination Act of 2008 (2008). *Public Law 110–233, 42 U.S.C. 2000ff et seq.*

Gionis, A., P. Indyk, and R. Motwani (1999a). *Similarity Search in High Dimensions via Hashing*. Proceedings of the 25th International Conference on Very Large Data Bases (VLDB), 518–529.

— (1999b). "Similarity Search in High Dimensions via Hashing". In: *Proceedings of the 25th International Conference on Very Large Data Bases*. Foundational locality-sensitive hashing reference; see also prior Datar et al. work already in the bibliography., pp. 518–529.

Gonzalez v. Trevino (2024). *602 U.S. 653*. Narrow construction of the Nieves exception.

Haber, S. and W. S. Stornetta (1991). *How to Time-Stamp a Digital Document*. Journal of Cryptology 3(2), 99–111.

Harlow v. Fitzgerald (1982). *457 U.S. 800*. Objective qualified-immunity standard.

Health Insurance Portability and Accountability Act of 1996 (1996). *Public Law 104–191, 42 U.S.C. 1320d et seq.*

Heck v. Humphrey (1994). *512 U.S. 477*. Favourable termination and the habeas bar.

Imbler v. Pachtman (1976). *424 U.S. 409*. Absolute prosecutorial immunity.

Individuals with Disabilities Education Act (2004). *20 U.S.C. 1400 et seq.* United States Code.

International Covenant on Civil and Political Rights (1966). *999 U.N.T.S. 171, entered into force 23 March 1976*.

Krawczyk, H. and P. Eronen (2010). *RFC 5869: HMAC-based Extract-and-Expand Key Derivation Function (HKDF)*. Tech. rep. Internet Engineering Task Force.

Kumho Tire Company v. Carmichael (1999). *526 U.S. 137*. Daubert factors extended to non-scientific experts.

Lane, T. v. (2004). "541 U.S. 509". In: *United States Reports* 541. Congress validly abrogated state sovereign immunity under Title II of the ADA as applied to access to the courts., p. 509.

Manuel v. City of Joliet (2017). *580 U.S. 357*. Pretrial detention without probable cause as a Fourth Amendment claim.

Maryland Rule 15-701 (n.d.). Maryland Rules of Procedure. Governs issuance of writs of mandamus and other extraordinary writs in the circuit courts.

Maryland v. King (2013). *569 U.S. 435*. DNA identification and the Fourth Amendment.

Mathews v. Eldridge (1976). *424 U.S. 319*. Three-factor procedural-due-process balancing test.

Md. Code Ann., Cts. & Jud. Proc. § 1-501 (n.d.). Maryland Courts and Judicial Proceedings Article. Circuit courts are courts of general equity jurisdiction.

Merkle, R. C. (1979). *A Certified Digital Signature*. Advances in Cryptology (CRYPTO '89), 218–238.

— (1980). “Protocols for Public Key Cryptosystems”. In: *Proceedings of the 1980 IEEE Symposium on Security and Privacy*, pp. 122–134.

Monell v. Department of Social Services (1978). *436 U.S. 658*. Municipal liability under 42 U.S.C. 1983.

National Institute of Standards and Technology (2019). *Security Requirements for Cryptographic Modules*. FIPS 140-3.

— (2020). *Zero Trust Architecture*. NIST Special Publication 800-207.

— (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1.

— (2024a). *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1.

— (2024b). *Module-Lattice-Based Digital Signature Standard*. FIPS 204 (ML-DSA).

— (2024c). *Module-Lattice-Based Key-Encapsulation Mechanism Standard*. FIPS 203 (ML-KEM).

— (2024d). *Stateless Hash-Based Digital Signature Standard*. FIPS 205 (SLH-DSA).

National Security Agency (2022a). *Commercial National Security Algorithm Suite 2.0 (CNSA 2.0)*. NSA Cybersecurity Advisory.

— (2022b). *Commercial National Security Algorithm Suite 2.0 (CNSA 2.0)*. NSA Cybersecurity Advisory.

Nieves v. Bartlett (2019). *587 U.S. 391*. Retaliatory arrest requires motive plus absence of probable cause.

Office of Management and Budget (2024). *Advancing Governance, Innovation, and Risk Management for Agency Use of Artificial Intelligence*. OMB Memorandum M-24-10.

Olmstead v. L.C. ex rel. Zimring (1999). *527 U.S. 581*. Integration mandate under ADA Title II.

Riley v. California (2014). *573 U.S. 373*. Warrant requirement for cell-phone searches.

Sanders, B. and J. B. Larson (2020). *Nearly 110,000 Died Awaiting Disability Benefit Appeals Between 2008 and 2019*. Joint congressional news release, based on GAO-20-641R.

Securities Exchange Act of 1934 (1934). *Section 13(f), 15 U.S.C. 78m(f)*. United States Code.

Shinei, C., Y. Masuyama, H. Abe, M. Miyakawa, T. Taniguchi, T. Ohshima, and T. Teraji (2025). “Homogeneous Spin-Dephasing Time of NV Centre in Millimetre-Scale ^{12}C-Enriched High-Pressure High-Temperature Diamond Crystals”. In: *Communications Materials* 6, p. 66.

Small Business Reorganization Act of 2019 (2019). *Public Law 116–54, 11 U.S.C. 1181 et seq. (Subchapter V)*.

Social Security Act (1935). *42 U.S.C. 301 et seq.* United States Code.

The White House (2023). *Executive Order 14110: Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence*. Revoked January 2025.

— (2025a). *Executive Order 14179: Removing Barriers to American Leadership in Artificial Intelligence*.

— (2025b). *Winning the Race: America’s AI Action Plan*.

Thompson v. Clark (2022). *596 U.S. 36*. Favourable termination requires only a termination without conviction.

U.S. Government Accountability Office (2020). *Social Security Disability: Information on Wait Times, Bankruptcies, and Deaths among Applicants Who Appealed Benefit Denials*. GAO-20-641R.

UN Human Rights Committee (2018). *General Comment No. 36 on Article 6 of the ICCPR: Right to Life, CCPR/C/GC/36*.

UN Human Rights Council (2007). *Institution-Building of the United Nations Human Rights Council, Res. 5/1*.

Uniform Directed Trust Act (2017). Uniform Law Commission. Model act separating trust direction and administration; adopted with variations in a growing number of states.

United States v. Little Lake Misere Land Company (1973). *412 U.S. 580*. Supremacy Clause and federal interests in land transactions.

USEC Inc., Case No. 14-10492 (Bankr. D. Del.) (2014). United States Bankruptcy Court for the District of Delaware. Chapter 11 proceedings; plan of reorganisation confirmed 30 September 2014; reorganized entity emerged as Centrus Energy Corp. (NYSE: LEU).

Van de Kamp v. Goldstein (2009). *555 U.S. 335*. Administrative acts and absolute immunity.

Winter v. Natural Resources Defense Council (2008). *555 U.S. 7*. Four-factor preliminary-injunction standard.

Younger v. Harris (1971). *401 U.S. 37*. Federal abstention from pending state criminal proceedings.

Zhuang, J.-Z., Y.-K. Wu, and L.-M. Duan (2024). “Hardware-Efficient Variational Quantum Algorithm in a Trapped-Ion Quantum Computer”. In: *Physical Review A* 110, p. 062414.

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